# Lab 2: Post Deployment Configuration

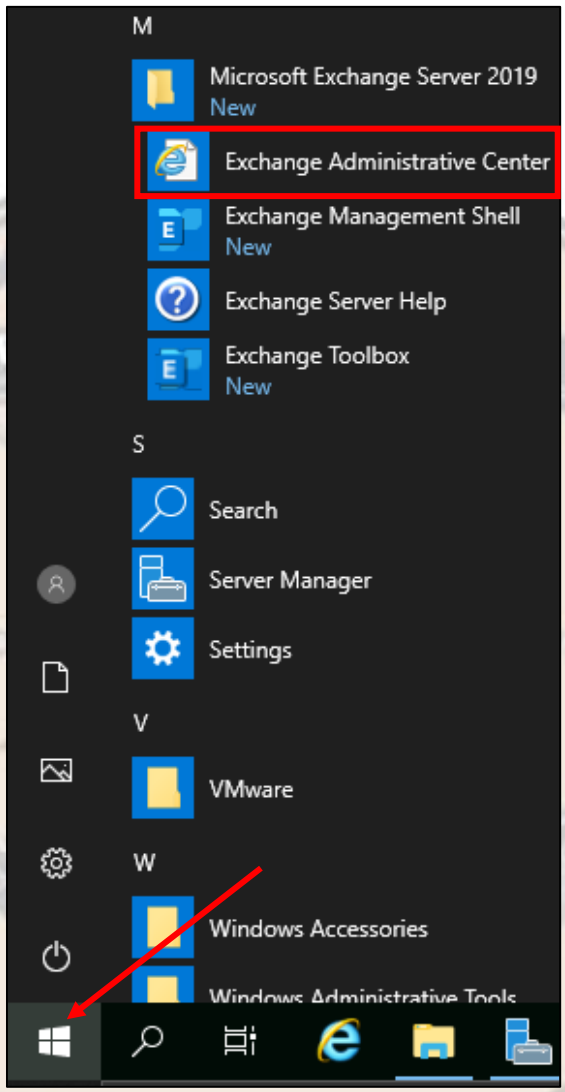
**Objective:** Continuing from Annex D, Lab 1. During this lab, the primary focus will be post installation configuration of Microsoft Exchange 2019 Mailbox Server Role. This lab consists of configuring your exchange environment to send and receive E-Mail, manage databases and log files, configure proper DNS settings to allow multiple instances of exchange. Configure access to email from internal and external sites, and create and manage Database Availability Groups (DAG).

Tasks:

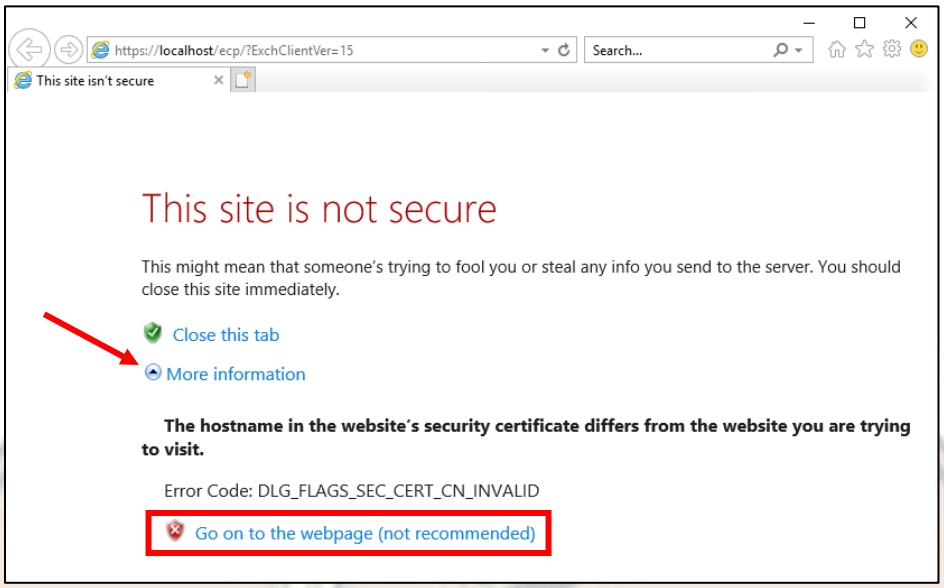
- **Rename and Move Databases and their Log Files.**
- **Finish Configuring Second Exchange Server**
- **Configure Mail Flow and Client Access**
- **Configure DNS for Exchange**
- **Configure Outlook Anywhere and Internal / External URLs**
- **Database Availability Groups**

## Task One: Rename Mailbox Database, Move Database and Move Log Files

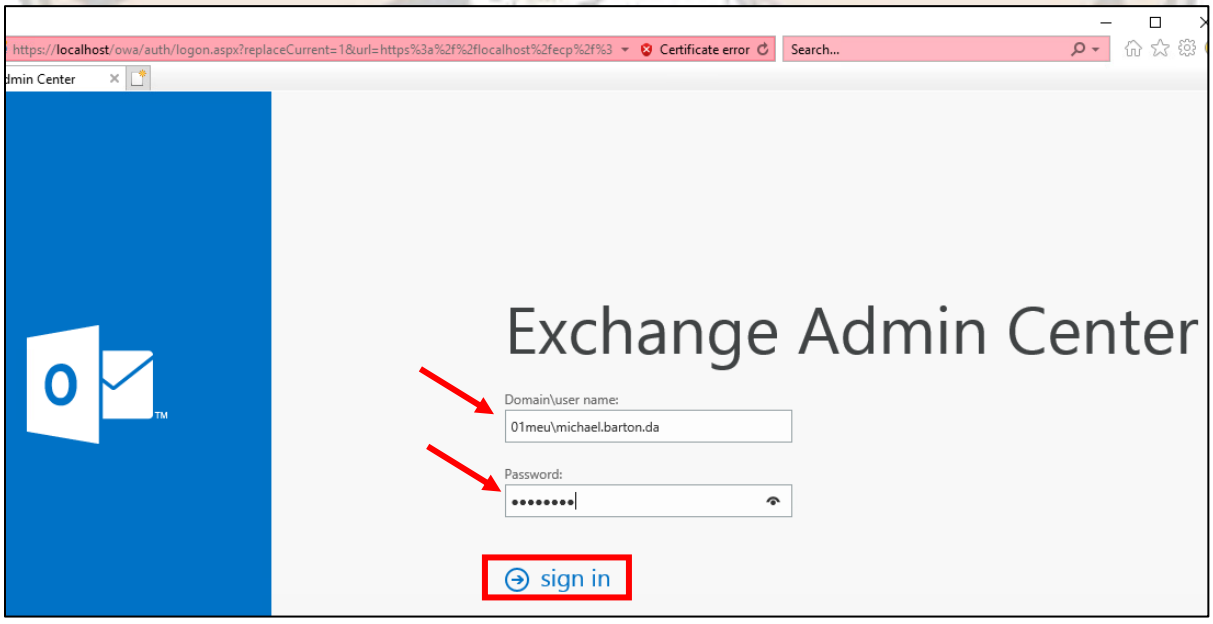
1. Log into the Exchange Administrative Center (**EAC**).



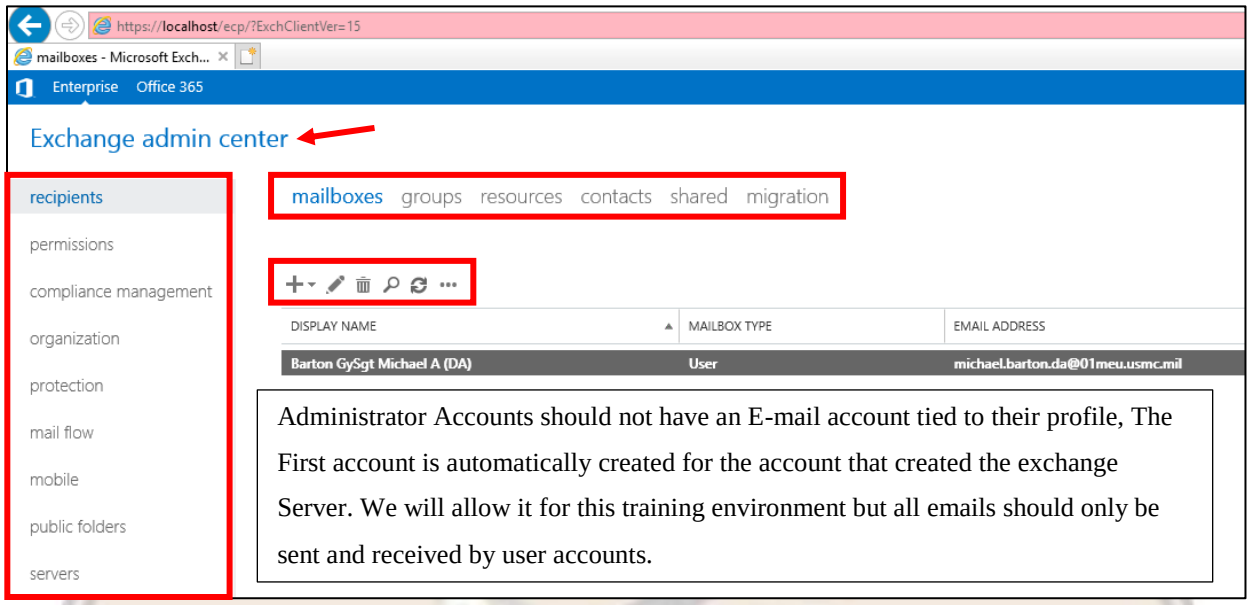
2. In the Browser, Expand **More Information**, then select **Go on to the webpage**.




3. Type in the **Domain\Username** and the **Password**, then press **Enter** or click on **Sign In**.



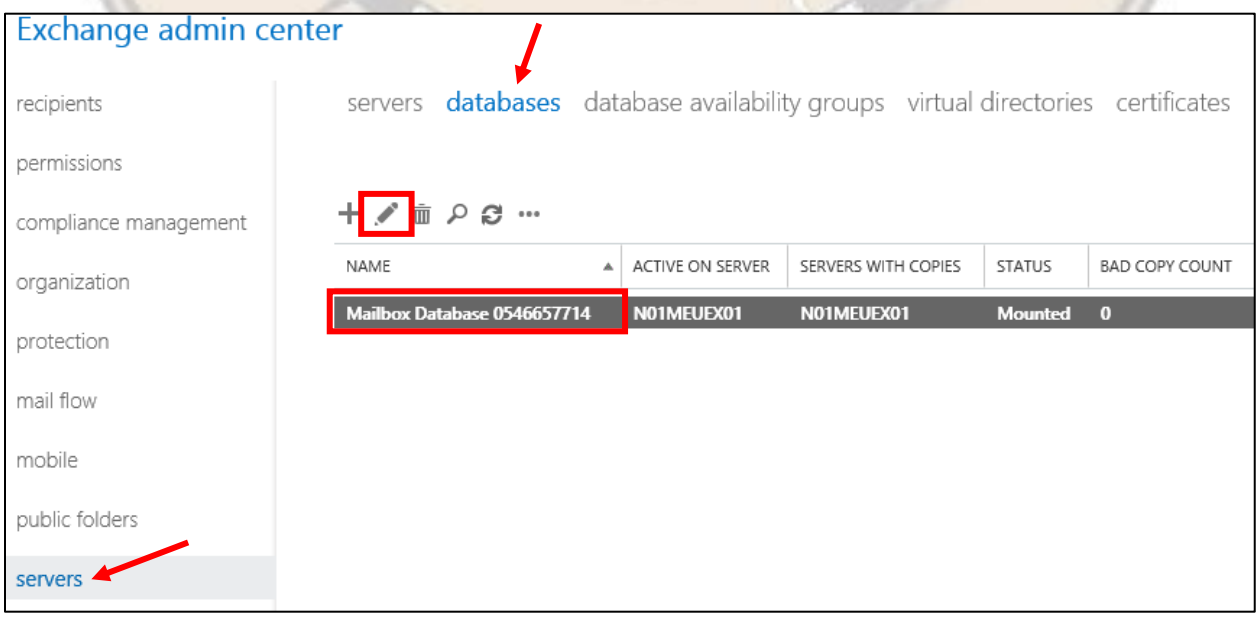
4. **You are now inside the EAC**. For more detailed information about each interface see the EAC Interface Elements Document in the **Lab Resources Folder**.



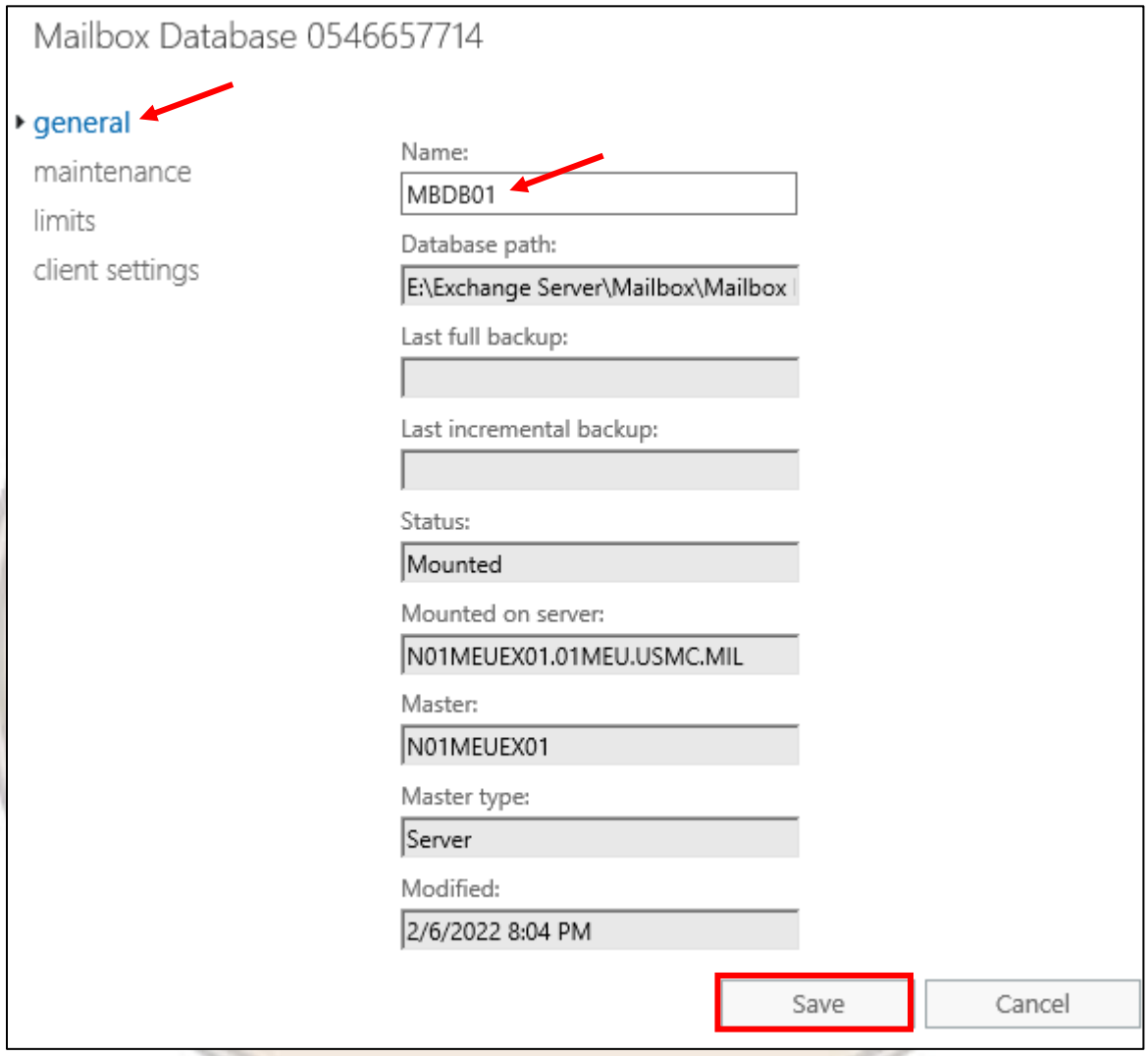

**5. Change Database file name from the EAC:**

A. Navigate to **Servers→Databases→** Select your Mailbox (MB) Database (DB), then Click on the Edit Icon .

**Note: As you can see the name is rather long which can increase the likelihood of typographical errors in subsequent steps.**



B. In the pop up, under the general tab change the name to match “MBDB01” for the first Exchange Server. The Second Server will be MBDB02. Click Save to implement the changes.




**6. Move Database (DB) and Transaction Log Files:**

A. By default the exchange DB and Transaction logs are placed on the system volume, however it is a recommended best practice to separate these onto separate discs so that the Mailbox DB is not affected by system I/O usage or system failure or OS disk corruption.

B. In NXXMEUEX01, Add Folders to create file paths for:

**F:\Exchange Database Files\MBDB01**

**F:\Exchange Database Files\MBDB02**

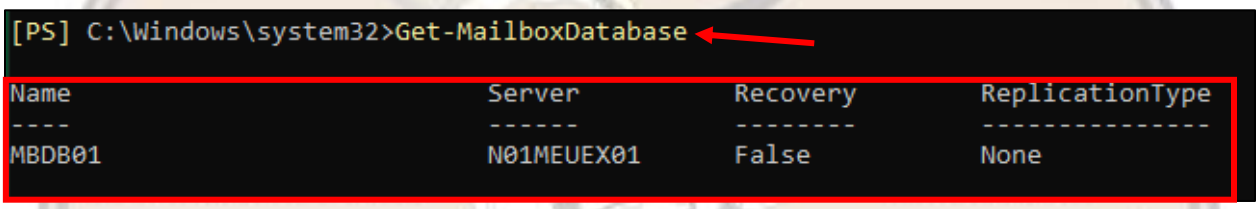
**F:\Exchange Log Files\MBDB01**

Note- When you configure XXMEUEX02 you will create the same file paths in it.

C. The database path and log path must be unique for each database on a single server. By the end, both exchange servers will host a duplicate copy of the others database.

D. Open the **Exchange Management Shell** as administrator and input:

**Get-MailboxDatabase** to ensure you are working with the correct Database.

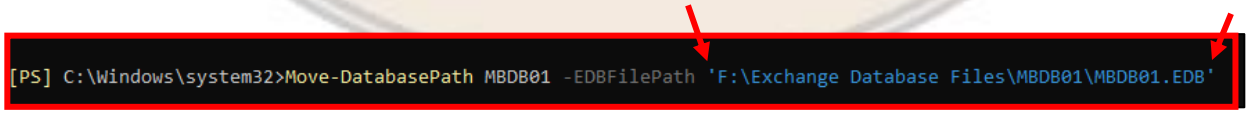



E. Once you confirm you are moving the correct Database utilize the

**Move-DatabasePath <Database Name> -EdbFilePath <New Path and FileName>**

Your output should look similar to this if you are moving your Database to the F: drive:

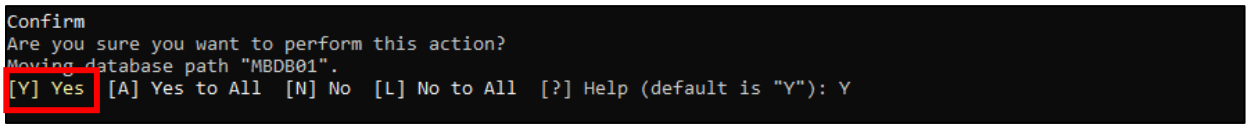
**Move-DatabasePath MBDB01 -EdbFilePath 'F:\Exchange Database Files\MBDB01\MBDB01.edb'**



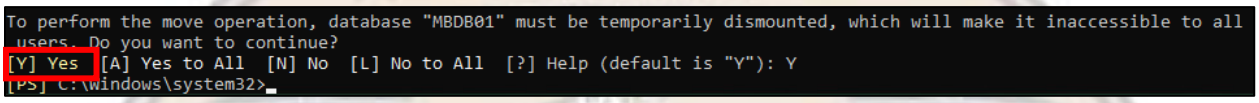

**NOTE-** If there is a space in the name of folder or document in the file path then the whole path must be encapsulated with ' ' or " ".

F. After pressing **Enter**, `EMS will ask Are you sure you want to perform this action [Y] [A] [N] [L] [?]`

Press **Y** and press **Enter**.




G. EMS then states that to perform this operation, database MBDB01 must be temporarily dismounted
Press Y and hit enter.




H. Keep in mind, your Database will be temporarily unavailable so in a live environment you would want to restrict this action to either initial set up, a fresh database not being used, or done during a scheduled maintenance window after hours.

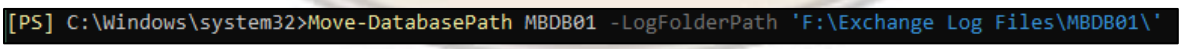
**7. Move Transaction Logs**

A. From the time an Exchange DB 'mounts' itself to a transaction log file *every* change made to the DB will be recorded in a log file before it is written log files are approximately 5 MB each and roll over to a new log file when they fill up. To move your transaction logs we will use the same command as above with a different switch: utilizing the –**LogFolderPath** to move your Log files to your third drive.

B. The paths for the log files must be different on each server.

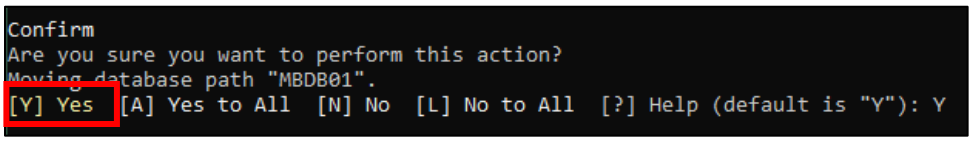
EXAMPLE:

```
Move-DatabasePath MBDB01 -LogFolderPath 'F:\Exchange Log Files\MBDB01\'
```



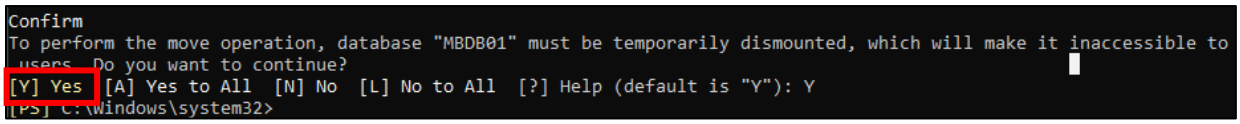

C. EMS will ask you if you are sure you want to perform this action

Press Y and hit enter



D. EMS will ask you to Confirm

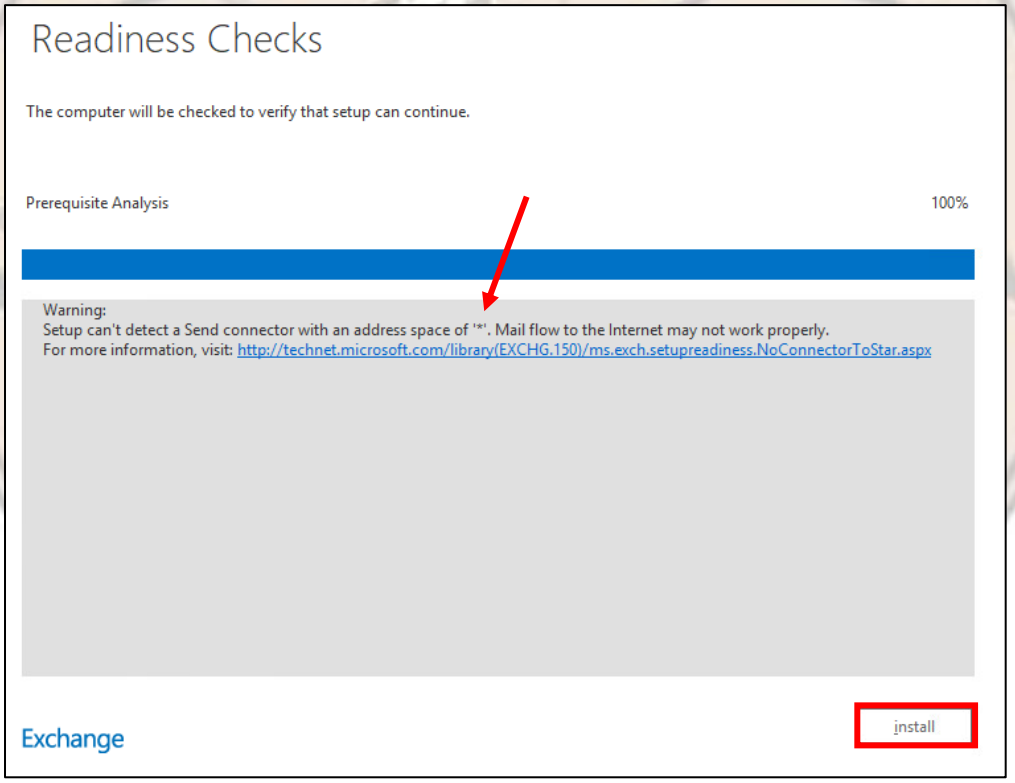
Press Y and hit enter




## Task Two: Finish Configuring Second Exchange Server

1. At this time, Finish Configuring NXXMEUEX02, bringing it to this point.

2. If you receive a warning about an asterisk ignore it and click Install.



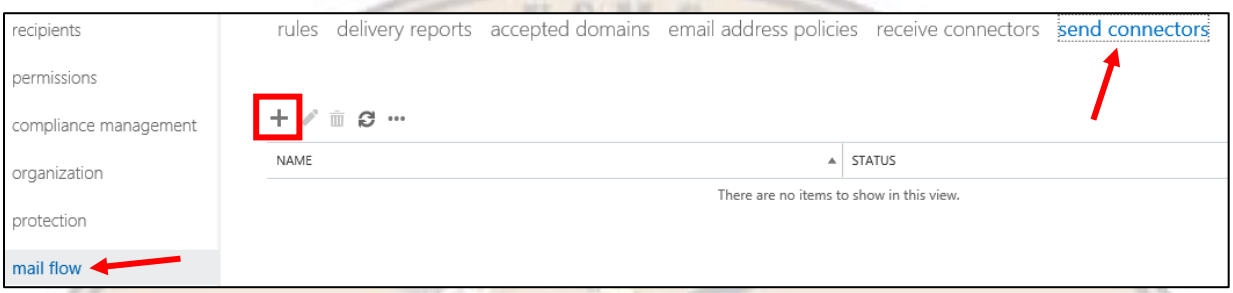

3. Once Exchange is installed and restarted, set up the folders on the F: drive just like you did on the first Exchange Server and move the Database and log files to the appropriate folders in the F: Drive.

## Task Three: Configure Mail Flow and Client Access

This can be done using either the Wizard or PowerShell but we will show you the Wizard. By default the receive connector and local accepted domains are created.

1. Use the EAC to create an internet Send connector. You can connect to the EAC from either exchange server

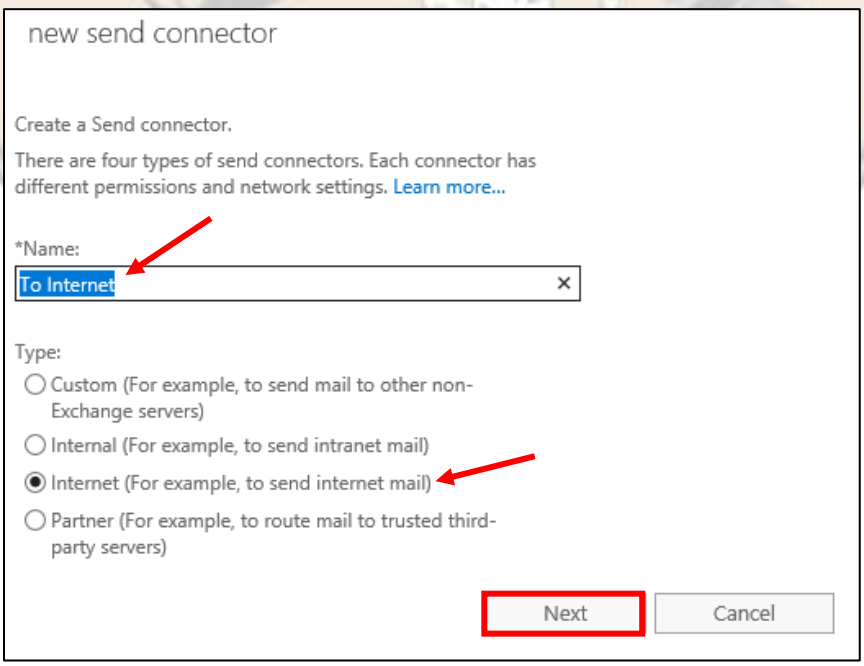
A. In the EAC, navigate to **Mail flow** > **Send connectors**, and then click **Add**+. This starts the **New Send connector** wizard.



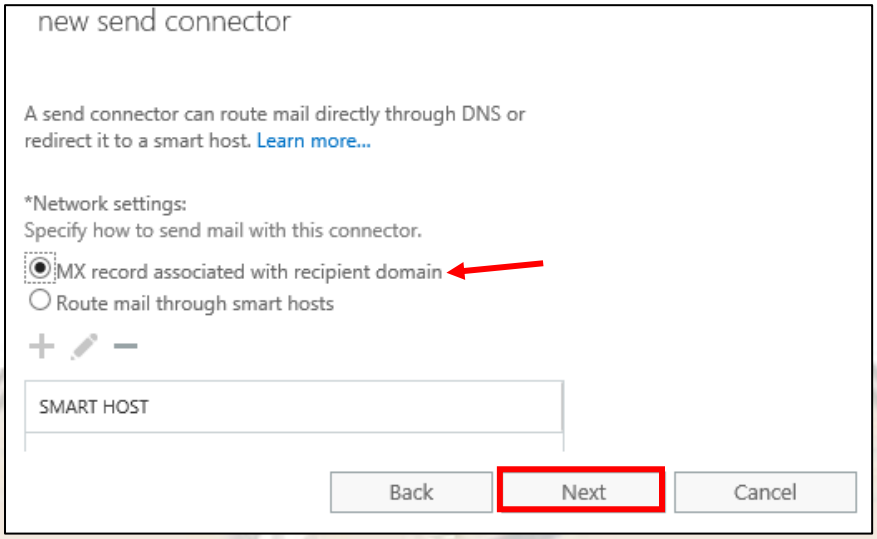

B. On the first page, enter the following information:

1) For the **Name**: Enter a descriptive name for the Send Connector (for example, **To Internet**).

2) For the **Type**: Select **Internet** then click **Next.**

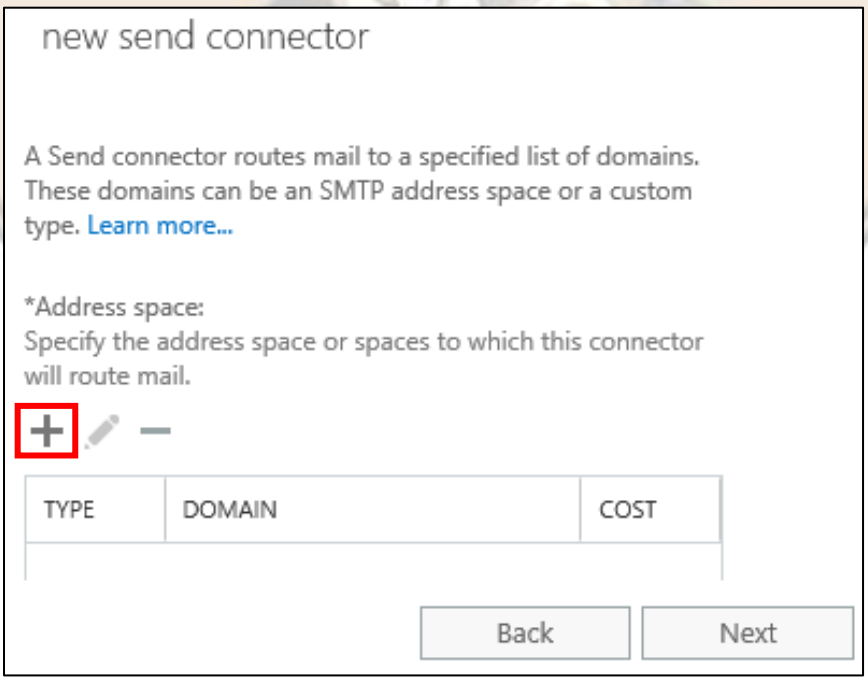


C. On the next page, verify that **MX record associated with recipient domain** is selected and click **Next.**

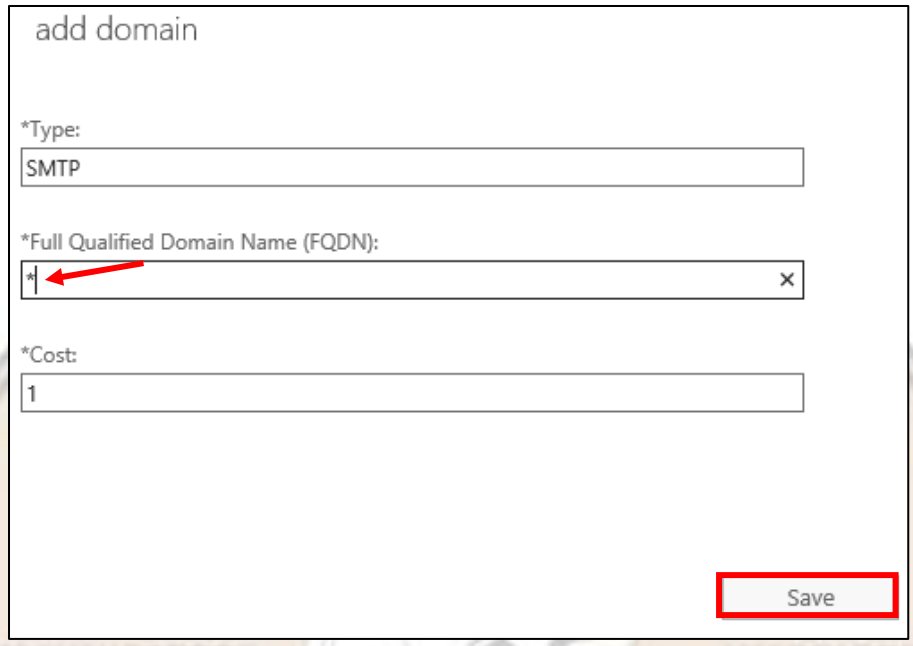



D. On the next page, enter the following information:

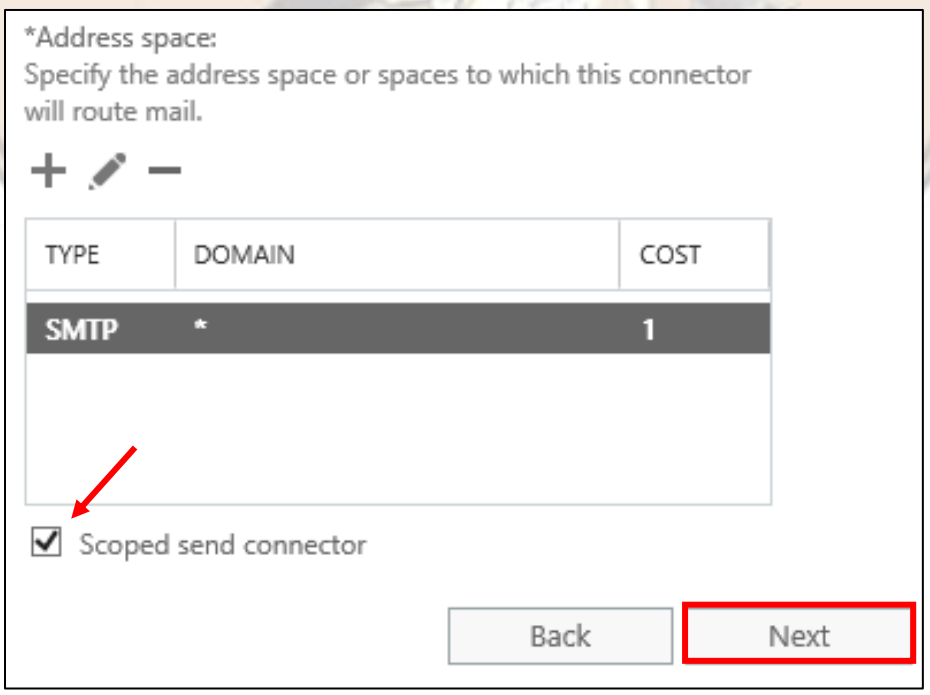
1) In the **Address space** section, click **Add** + .



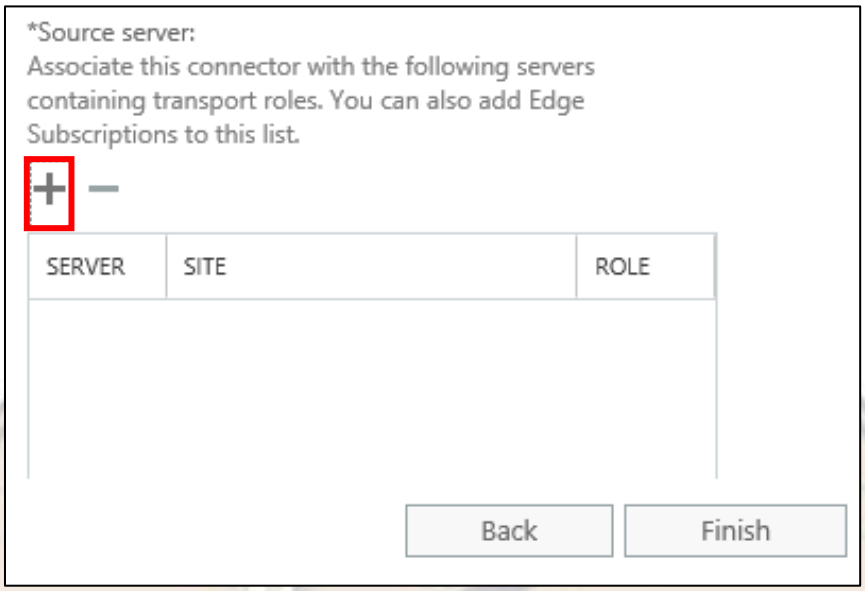
2) In the **Add domain** dialog box, in **Fully Qualified Domain Name (FQDN)**, enter an asterisk (*****), and then click **Save**. This asterisk value indicates that the Send connector applies to messages addressed to all external domains.




2) In **Scoped send connector** select **Scoped send connector**, since the connector will only be usable by the mailbox servers in the same Active Directory site. Click **Next.**

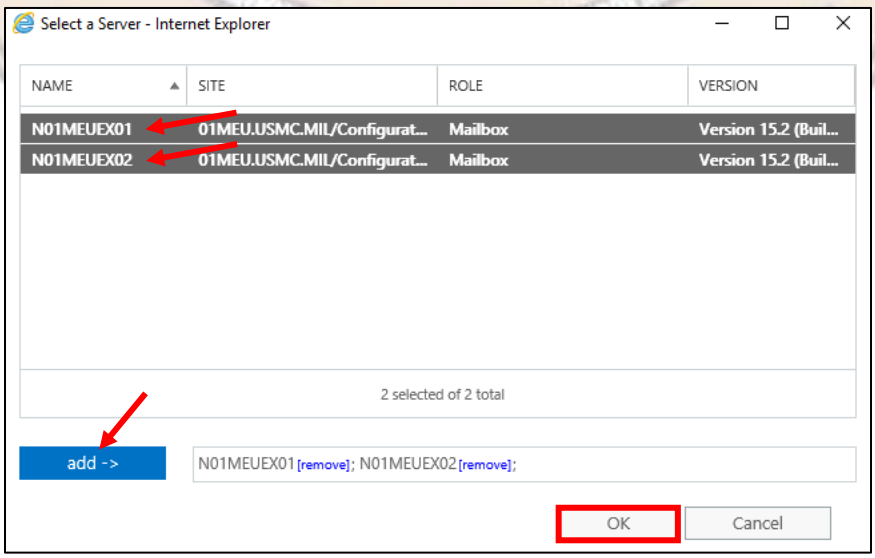


E. On the next page, in the **Source server** section, click **Add** +.

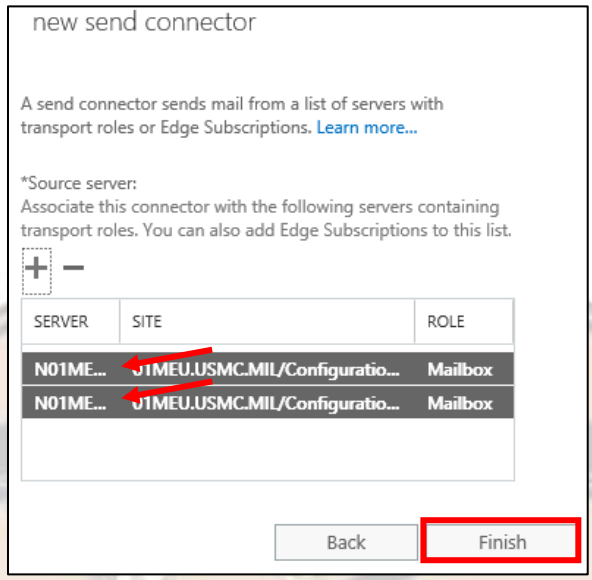



1) In the **Select a Server** dialog box that appears, select one or more Mailbox servers (select **BOTH** servers), that you want to use to send mail to the internet. If you have multiple Mailbox servers in your environment, select the ones that can route mail to the internet. If you have only one Mailbox server, select that one. After you've selected both Mailbox servers, click **Add**, then click **OK**.

CAPSTONE NOTE- In capstone you will only be configuring an additional exchange server in an organization. The * internet send connector will already exist, you just have to **add another source server** to it.



2) Both Exchange Servers are added, click **Finish**.




**F. Alternatively, (IF you want to)** you can try to use the Exchange Management Shell (EMS), to create the Internet Send Connector instead of EAC. If you want to try this then delete the send connector from EAC and use EMS.

1. Open the Exchange Management Shell and Use the following syntax:

```
New-SendConnector -Name <Name> -AddressSpaces * -Internet [-
   SourceTransportServer <fqdn1>,<fqdn2>...]
```

**EXAMPLE:**

```
New-SendConnector -Name To Internet -AddressSpaces * -Internet -
   SourceTransportServer XXMEUEX01.USMC.MIL, XXMEUEX02.USMC.MIL
```

2. Check EAC to make sure it was done right.

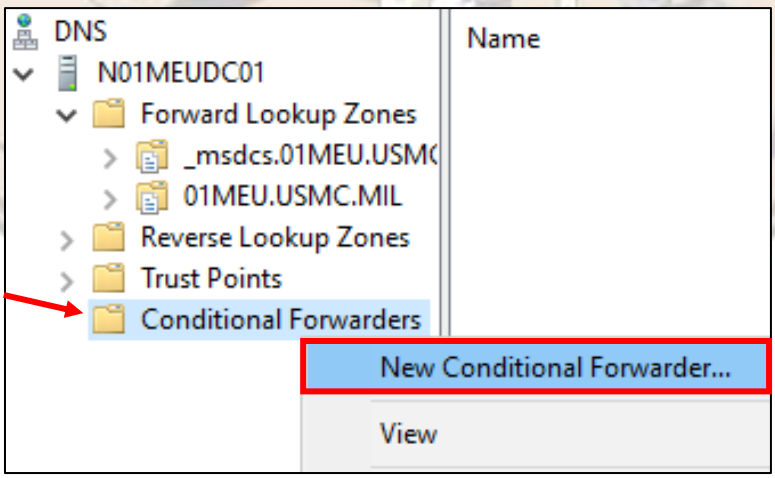
## Task Four: Configure DNS Settings to Support Exchange

Like many aspects of I.T., there are many configurations that can be done in other ways. One of those other ways is to use Conditional Forwarders instead of stub zones. Conditional Forwarders are neither better nor worse and are very similar to stub zones, but not as much information is shared between you and the distant site. Both the stub zone and conditional forwarders are forward lookup zones for a distant domain, meaning that they associate the domain name to the IP address. Every network should also have an associated Reverse Lookup Zone to translate IP addresses to Domain Names. **You cannot have a Stub Zone and a Conditional Forwarder for the same distant site/distant network.**

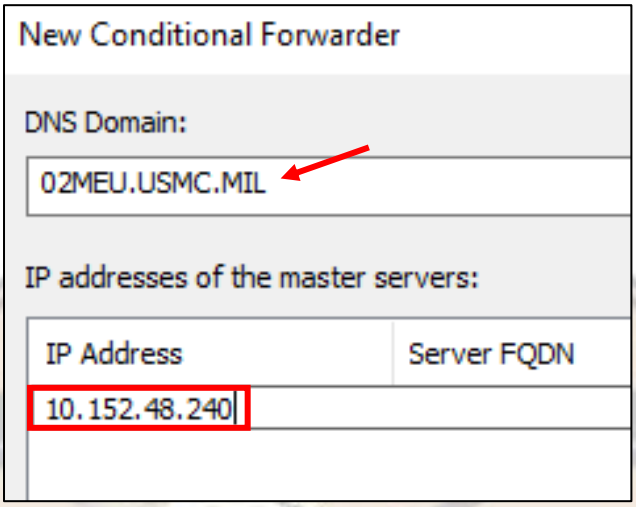
On your DNS Server **open the DNS manager** via the Server Manager →Tools→DNS:

1. **Conditional Forwarders**- You want to create a conditional forwarder or a Stub Zone, to any distant site that you want to communicate with (Send E-Mail and other types of communication). Create the zones as well as a reverse lookup zone for each site.

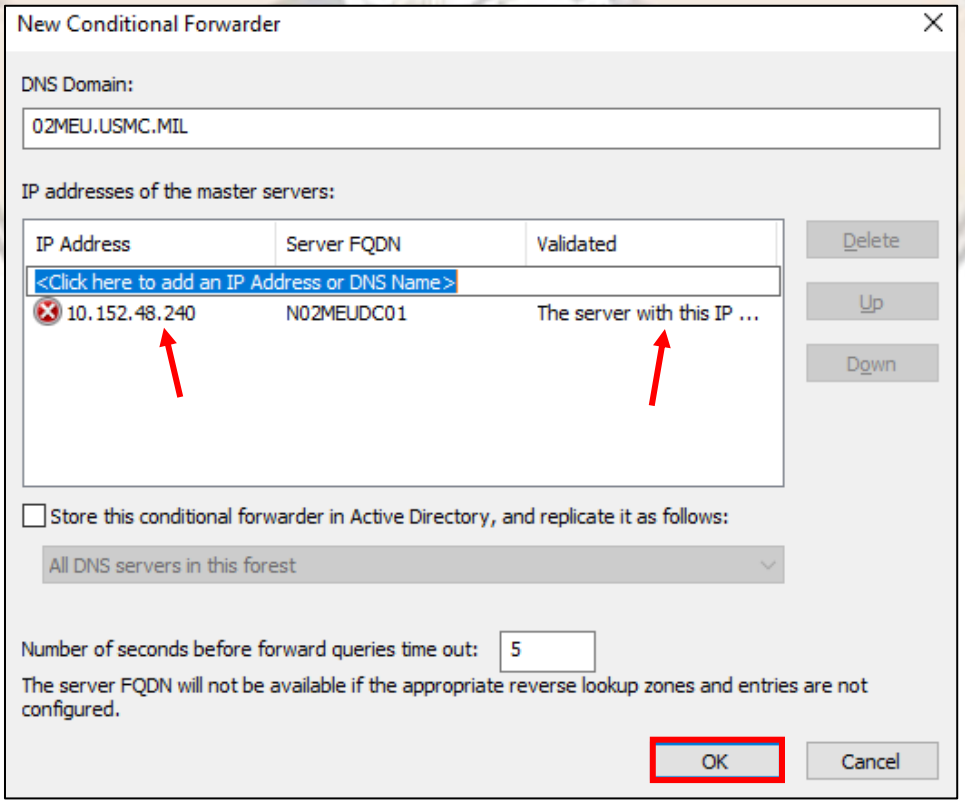
A. Under XXMEUDC01 Expand "**Conditional Forwarders**" → Right Click and select **New Conditional Forwarder**.



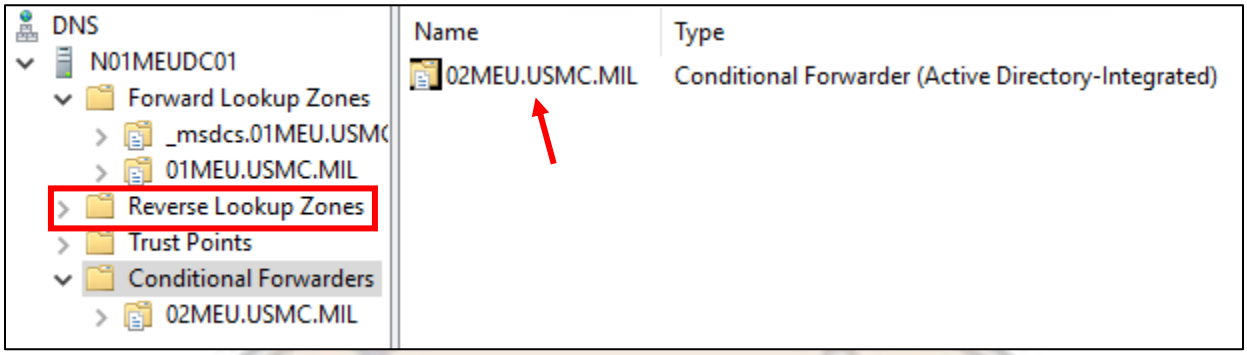
B. Type the target domain (Distant Domain) and its DNS server IP address, then Press Enter.



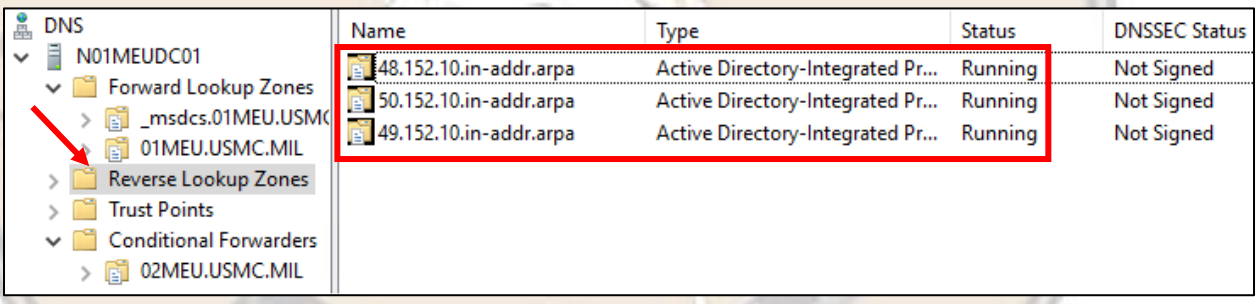

C. The distant domains Reverse lookup Zone needs to be created for Full Validation (Forward and Reverse). Click **OK**.



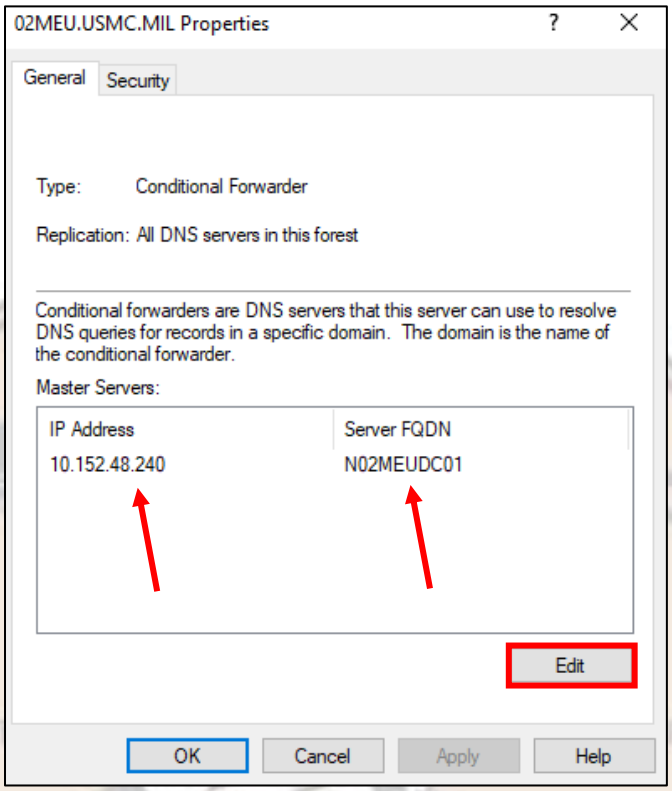
D. The Conditional Forwarder is set up, Click on Reverse Lookup Zones.



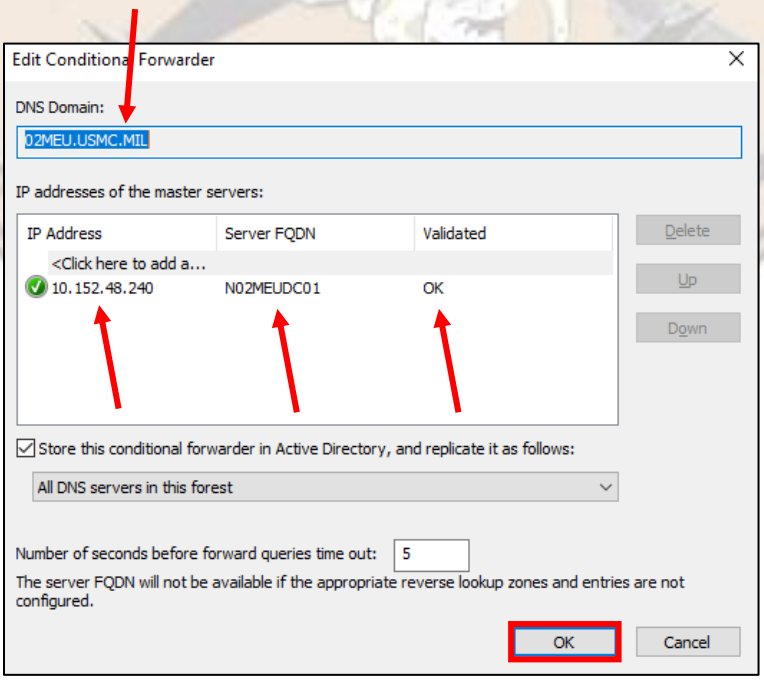

E. Create the Reverse lookup zones for your neighboring domains and your instructor's domain.



F. Return to your Conditional Forwarders, everything is resolving if both sides are properly configured. Click **Edit** to look at the other settings.




G. Again, everything is working as it should. Close the conditional Forwarders.



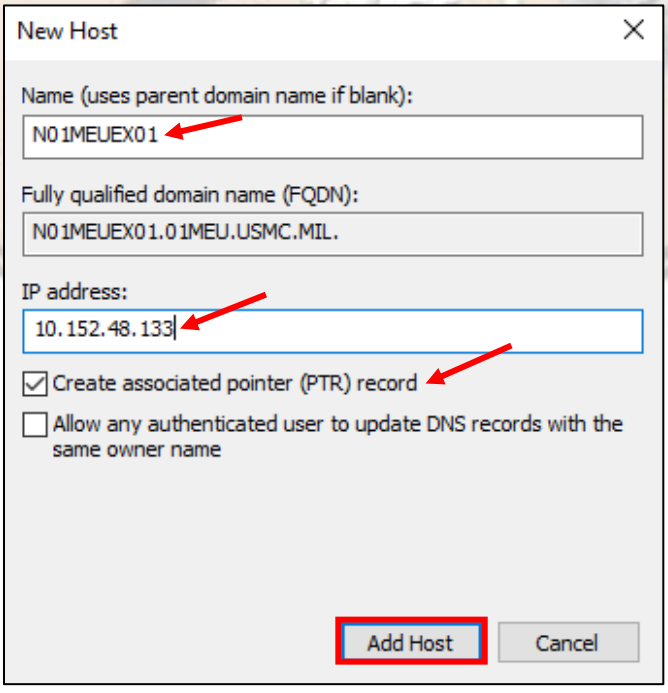
H. From this point your DC will be able to communicate with distant DC's that you have Conditional Forwarders set up with. (Both parties must have a conditional Forwarder set to each other and their own MX Record on their DNS server pointing to their Exchange server. There are a few more things that need to happen before you can send E-Mails.

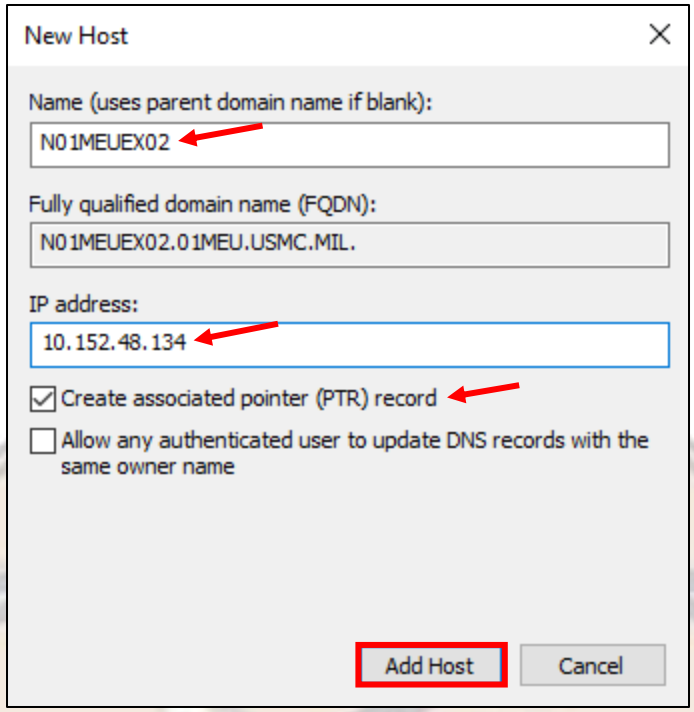
2. Configure DNS records.

| DNS RECORDS | | |
|---|---|---|
| **FQDN** | **DNS RECORD TYPE** | **VALUE** |
| NXXMEUEX01.XXMEU.USMC.MIL | A | 10.154.XXX.YYY (EX01) |
| NXXMEUEX02.XXMEU.USMC.MIL | A | 10.152.XXX.YYY (EX02) |
| AUTODOSCOVER.01MEU.USMC.MIL | A | 10.154.XXX.YYY (EX01) |
| AUTODOSCOVER.01MEU.USMC.MIL | A | 10.152.XXX.YYY (EX02) |
| WEBMAIL.XXMEU.USMC.MIL | A | 10.154.XXX.YYY (EX01) |
| WEBMAIL.XXMEU.USMC.MIL | A | 10.152.XXX.YYY (EX02) |
| OWA.XXMEU.USMC.MIL | A | 10.154.XXX.YYY (EX01) |
| OWA.XXMEU.USMC.MIL | A | 10.152.XXX.YYY (EX02) |
| XXMEU.USMC.MIL | MX | NXXMEUEX01.XXMEU.USMC.MIL |
| XXMEU.USMC.MIL | MX | NXXMEUEX02.XXMEU.USMC.MIL |
| Service: _autodiscover<br>Protocol: _tcp<br>Port Number: 443 | SRV | NXXMEUEX01,NXXMEUEX02 |

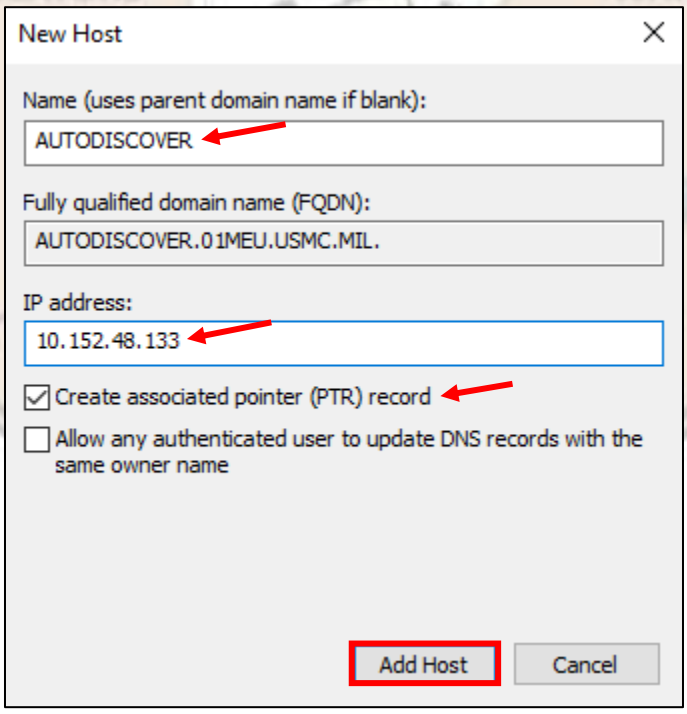
**A. Create Host-A and PTR** records from the table. **IF NOT CREATED ALREADY.**

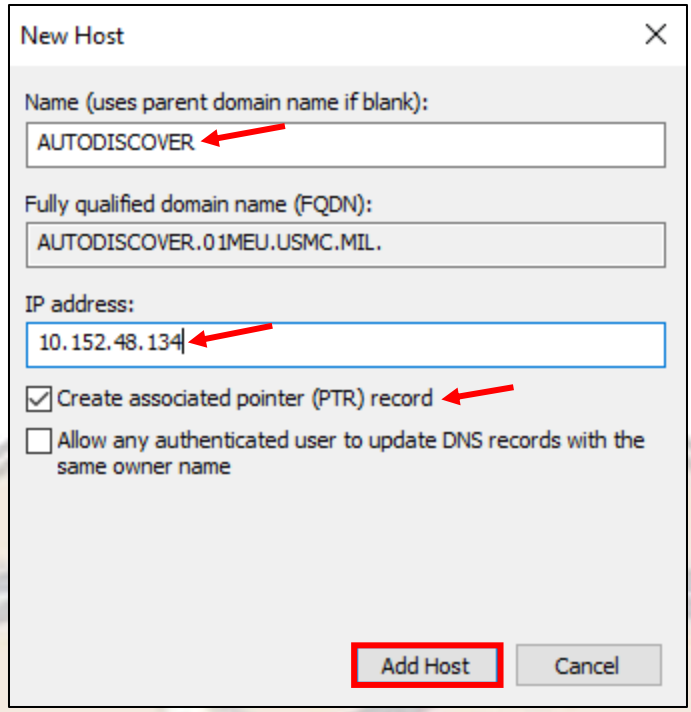
**1. Exchange Server Host Records for the VM's (Should already exist)**



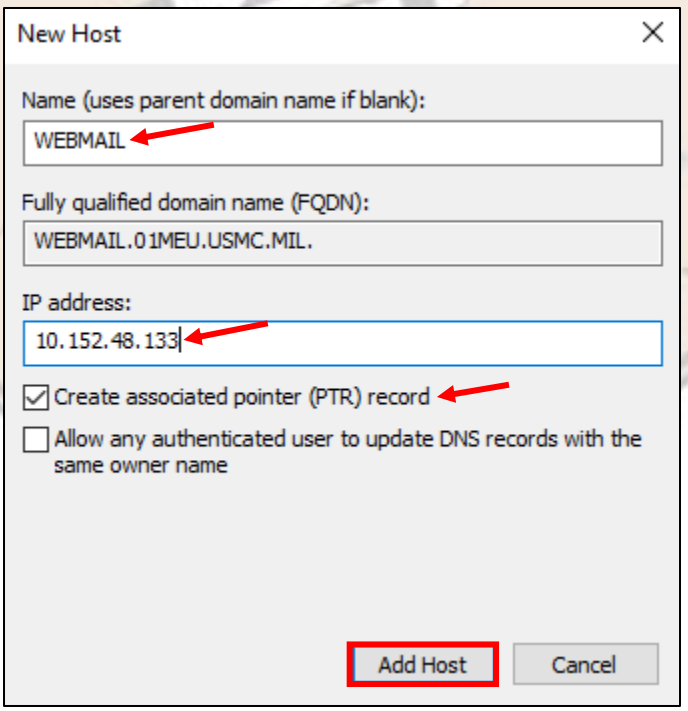


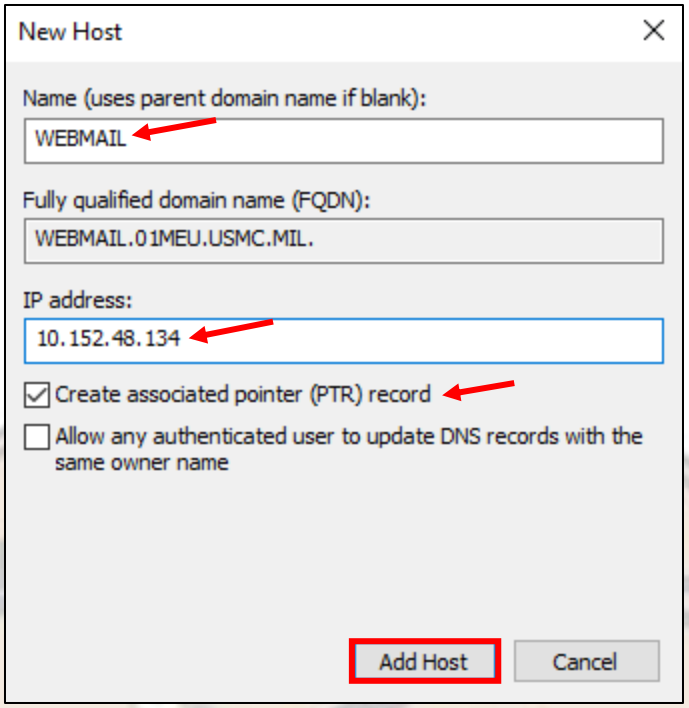

**2. Create Host-A records** named **AUTODISCOVER.USMC.MIL** that points to the IP address of the first and then second exchange server.



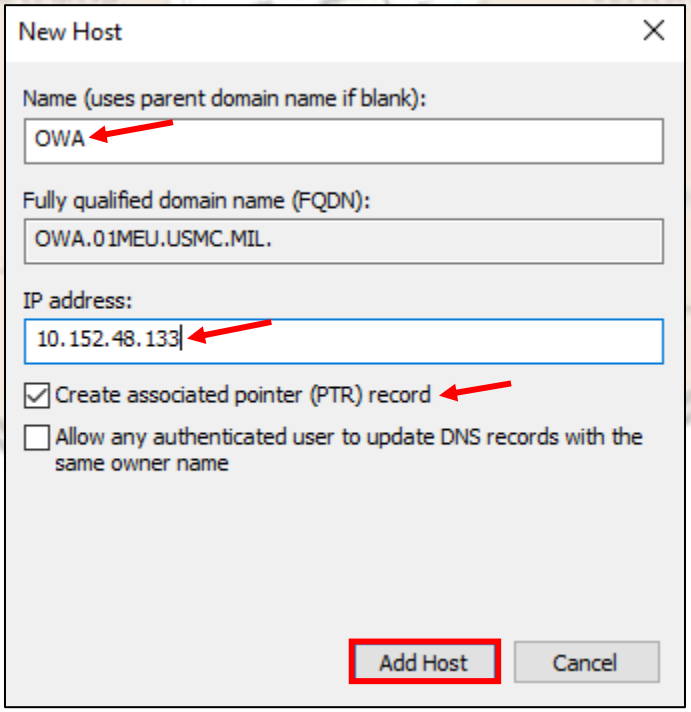


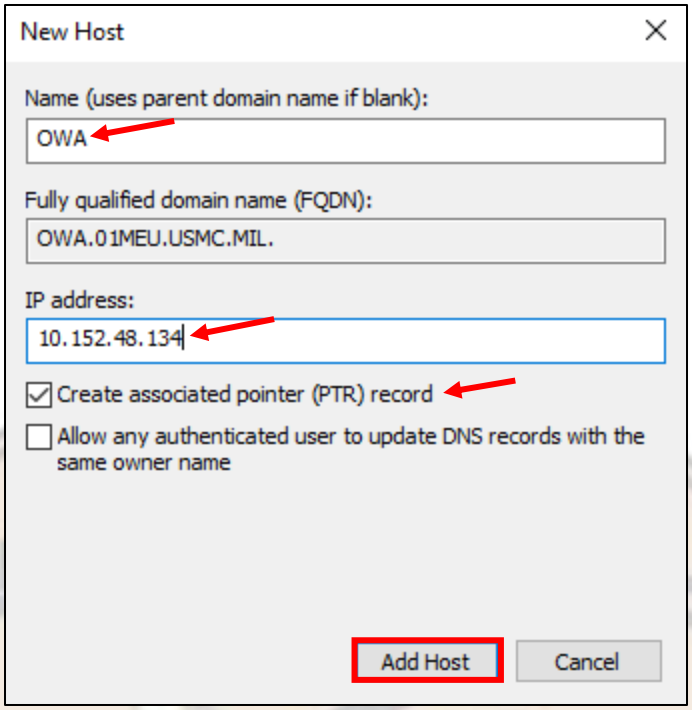

**3. Create Host-A records** named **MAIL.USMC.MIL** that points to the IP address of the first and then second exchange server.





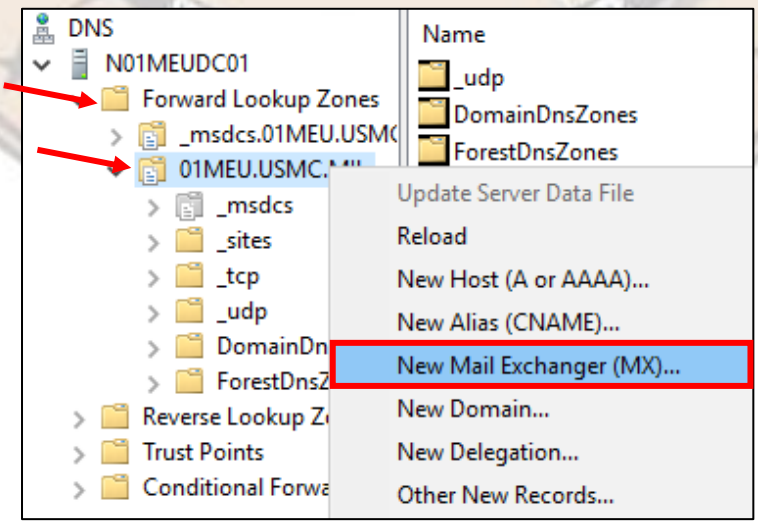

**3. Create Host-A records** named **OWA.USMC.MIL** that points to the IP address of the first and then second exchange server.



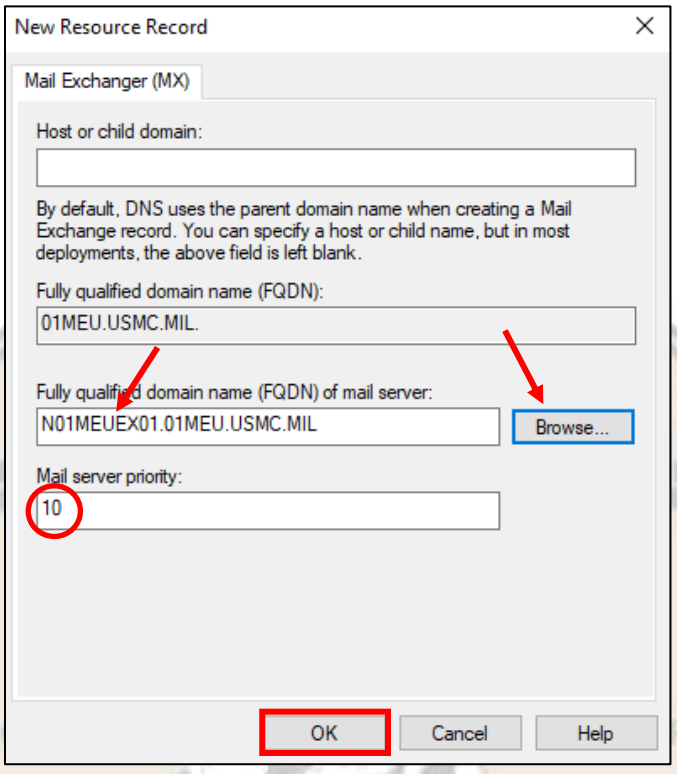


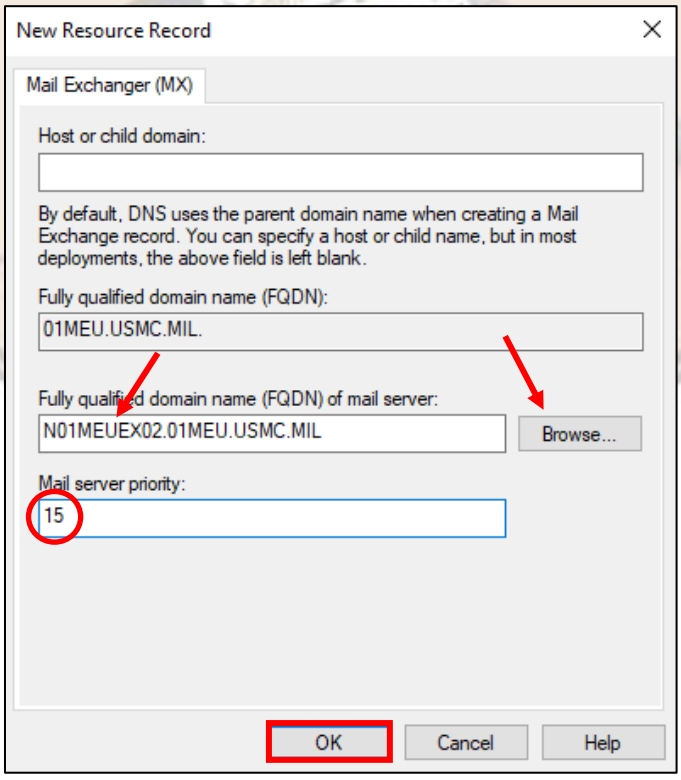

**B. Create MX Records for each Exchange Mailbox Role Server.**

1. Under XXMEUDC01 Expand "**Forward Lookup Zones**" → Expand **XXMEU.usmc.mil** → Right Click your domain and select "**New Mail Exchanger (MX)…**"



2. In the Fully Qualified Domain Name box, **input the <u>Mailbox</u> Server Host-A record**: MAIL.XXMEU.USMC.MIL.

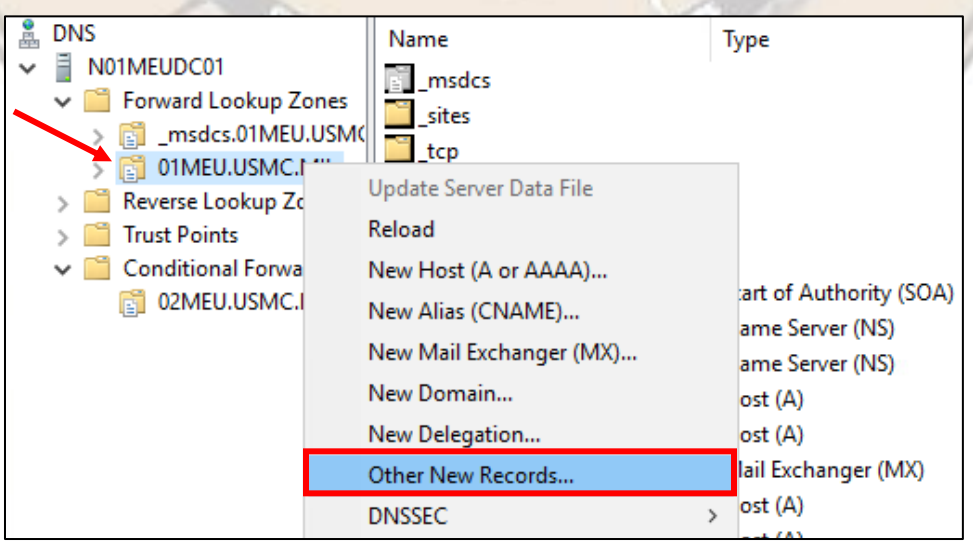


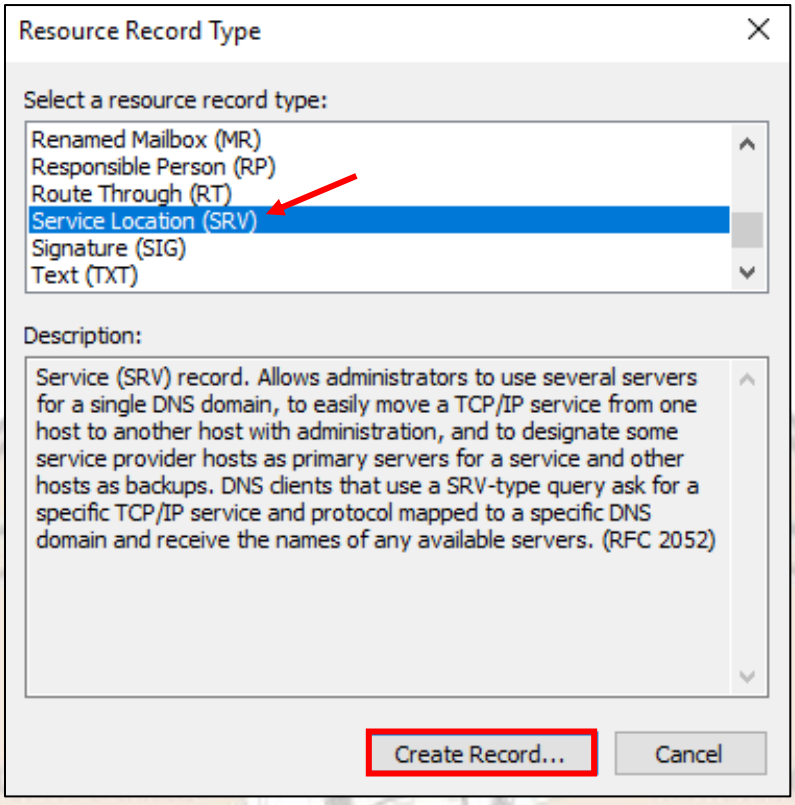
| | | |
|---|---|---|
| N01MEUEX01 | Host (A) | 10.152.48.133 |
| N01MEUEX02 | Host (A) | 10.152.48.134 |
| N01MEUFS01 | Host (A) | 10.152.48.138 |
| n01meupub | Host (A) | 10.152.48.141 |
| N01MEUUTIL | Host (A) | 10.152.48.140 |
| N01MEUVCSA01 | Host (A) | 10.152.48.153 |
| N01MEUWKS01 | Host (A) | 10.152.48.250 |
| N01MEUWKS02 | Host (A) | 10.152.48.251 |
| AUTODISCOVER | Host (A) | 10.152.48.133 |
| AUTODISCOVER | Host (A) | 10.152.48.134 |
| WEBMAIL | Host (A) | 10.152.48.133 |
| WEBMAIL | Host (A) | 10.152.48.134 |
| OWA | Host (A) | 10.152.48.133 |
| OWA | Host (A) | 10.152.48.134 |
| (same as parent folder) | Mail Exchanger (MX) | [10] N01MEUEX01.01MEU.USMC.MIL |
| (same as parent folder) | Mail Exchanger (MX) | [15] N01MEUEX02.01MEU.USMC.MIL |

**E. Configure the Service Record (SRV) for autodiscover.**

1. **Right Click** on the **Domain** under the <u>forward lookup Zone</u> and select **Other New Records…**



2. Select Service Location (SRV), then Click Create Record.




3. In the Service Location box enter the following inputs:

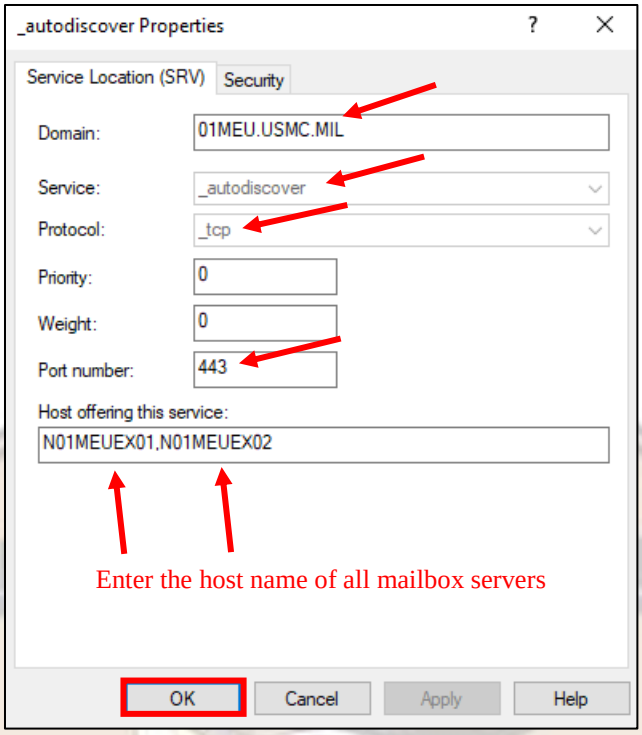
**Domain:** XXMEU.USMC.MIL

**Service:** _autodiscover

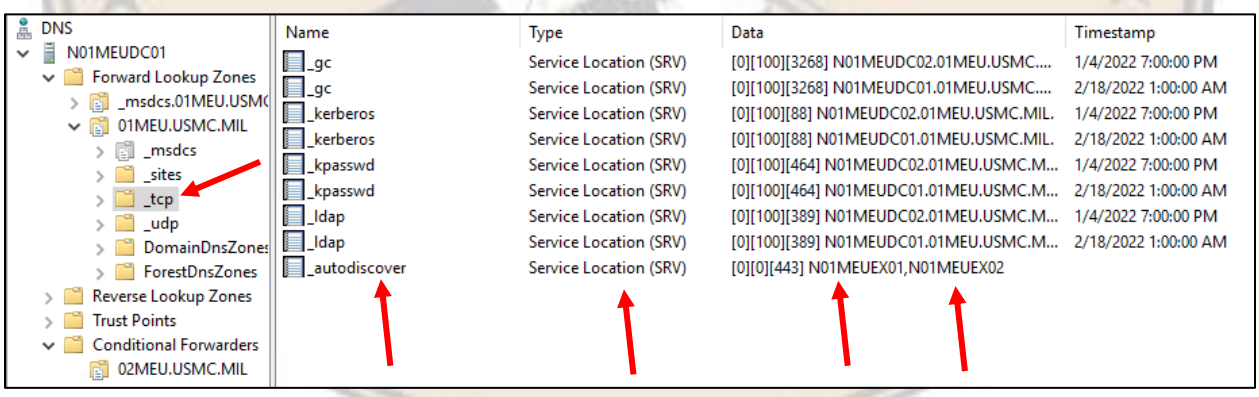
**Protocol:** _tcp

**Port number:** 443

**Host offering this service:** NXXMEUEX01,NXXMEUEX02




4. The new Service Location Record (SRV) has been added.




5. If you are building multiple mailbox role servers, you will have a Host-A record named MAIL.XXMEU.USMC.MIL for each mailbox server IP address. This results in the DNS server automatically forwarding requests for the Exchange mailbox server resources to the next mailbox server listed in DNS by using a process called DNS Round Robin (AKA Poor Man's Load Balancing).

6. Confirm your DNS Settings work correctly for Round Robin:

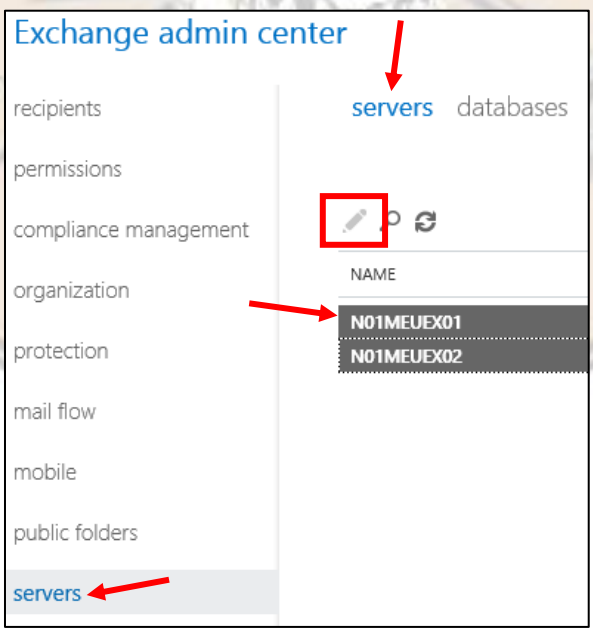
a. Open a web browser and attempt to navigate to the internal site URLs.

b. In Command Prompt or PowerShell, conduct an NSLOOKUP, three times for MAIL.XXMEU.USMC.MIL. Each time the domain controller will rotate the order of the exchange server list.

## Task Five: Configure Outlook Anywhere and Internal/External URL's.

A. **Outlook Anywhere** assists the **Microsoft Outlook Email Client** Software from the Microsoft Office Suite to connect to the exchange Server without administrator intervention.

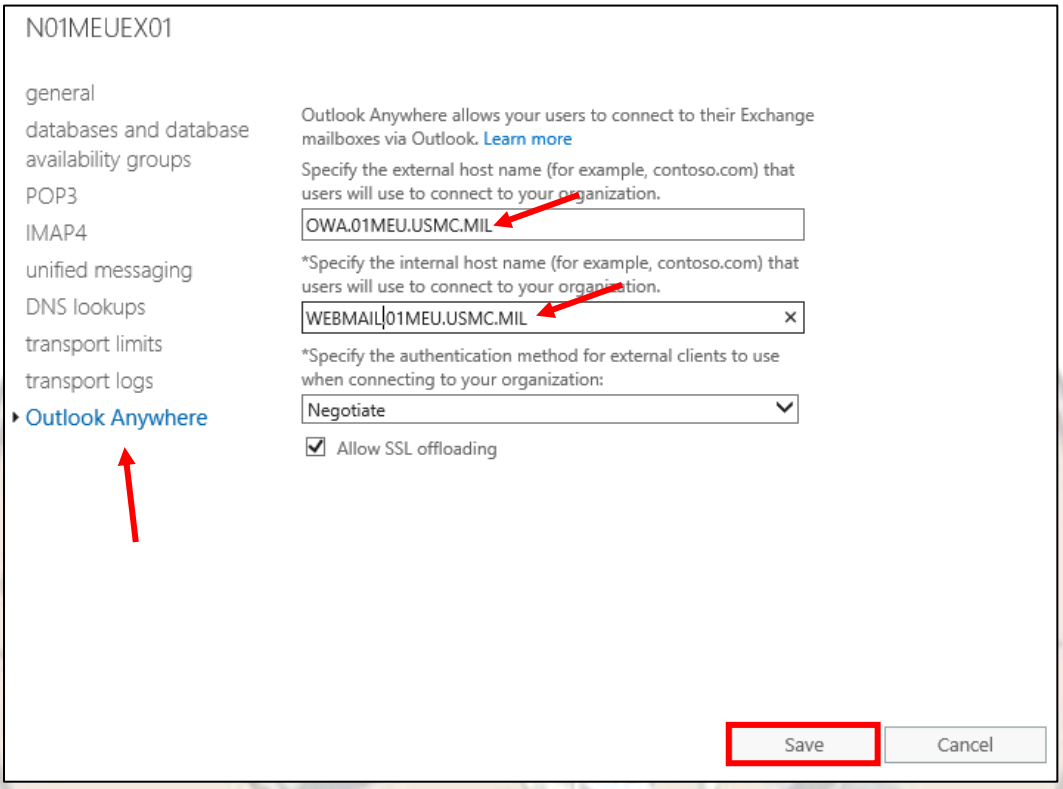
1. Open EAC → Servers → Servers, Select the first exchange server, then click on .



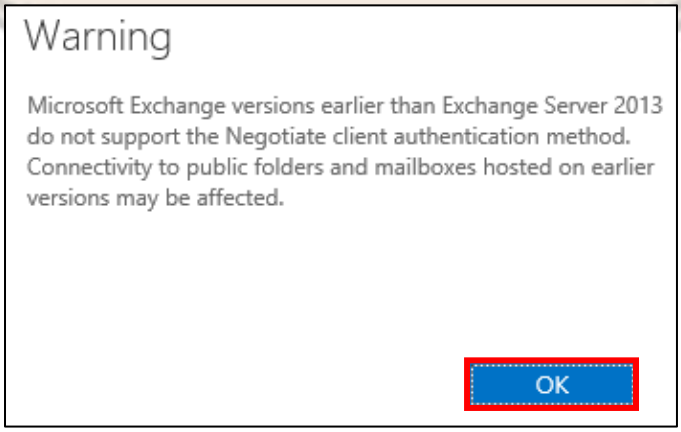
2. Click on **Outlook Anywhere**, then specify your **External and Internal host names** (Website you want users to go to from outside or inside of your domain), then click **Save.**

External: **OWA.XXMEU.USMC.MIL**

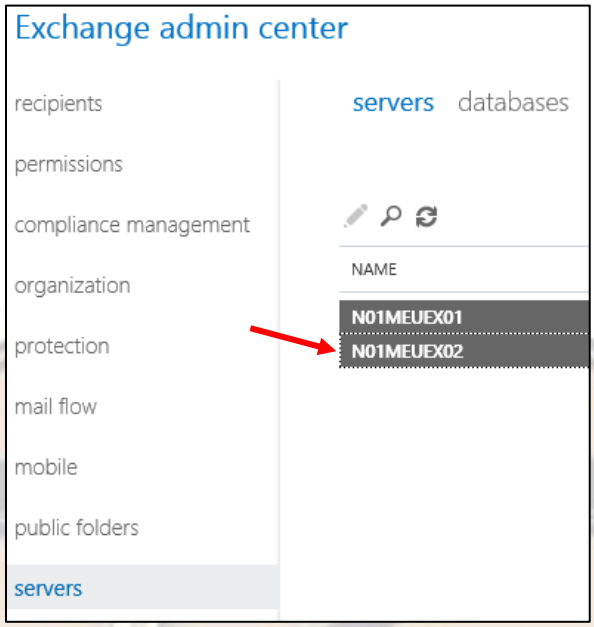
Internal: **WEBMAIL.XXMEU.USMC.MIL**




3. You do not have earlier versions of exchange, click **OK.**

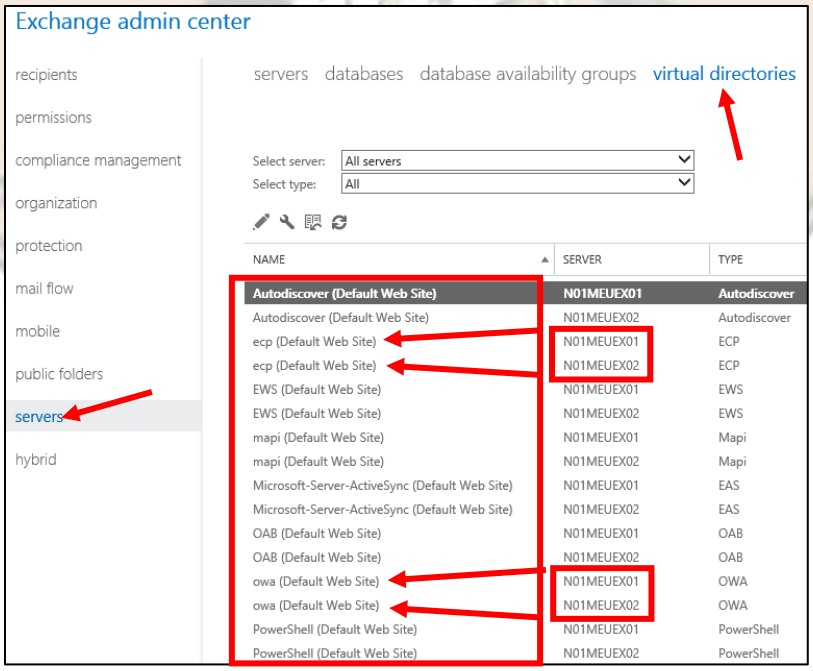


4. Complete steps 1-3 for the second Exchange Server




B. Configure **Virtual Directory - Internal and External URL's**.

1. Navigate to **Servers → Virtual directories.** Click on the services for each mailbox role server and configure the internal and external URL.

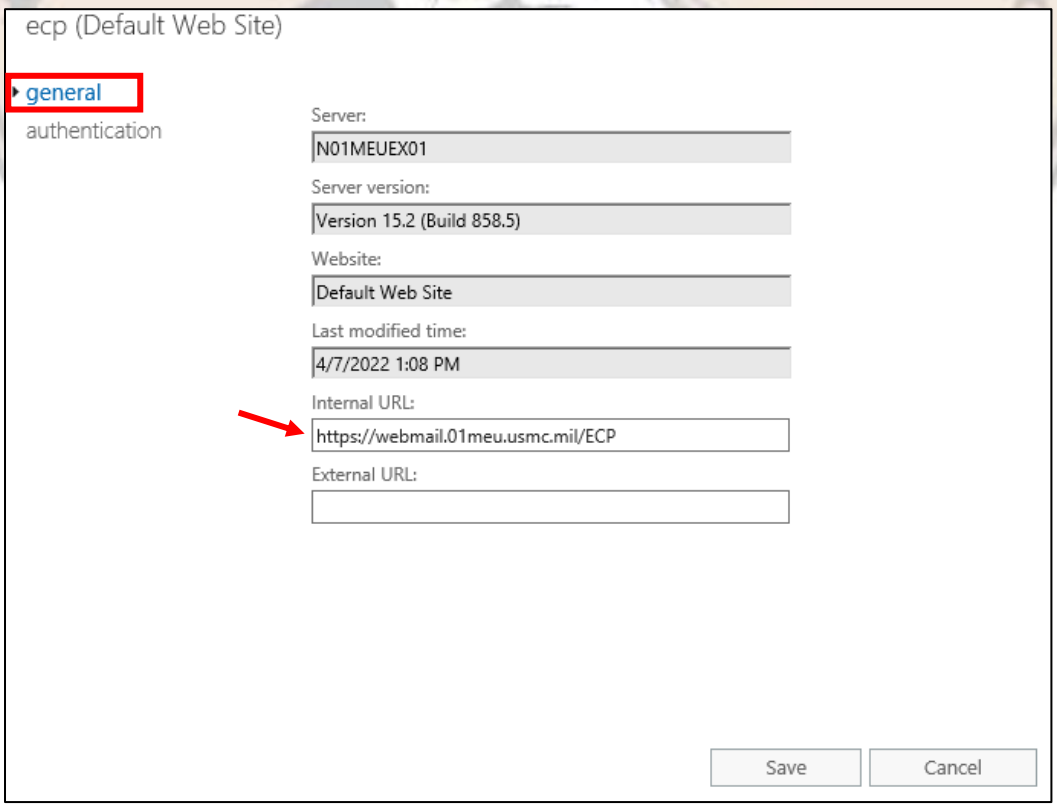


2. Use the following URLs for the internal and External URL's on each Exchange Mailbox Server. Some will not be changed and some cannot be changed. Some of these you do not want to make available externally, like ECP (EAC) and PowerShell.

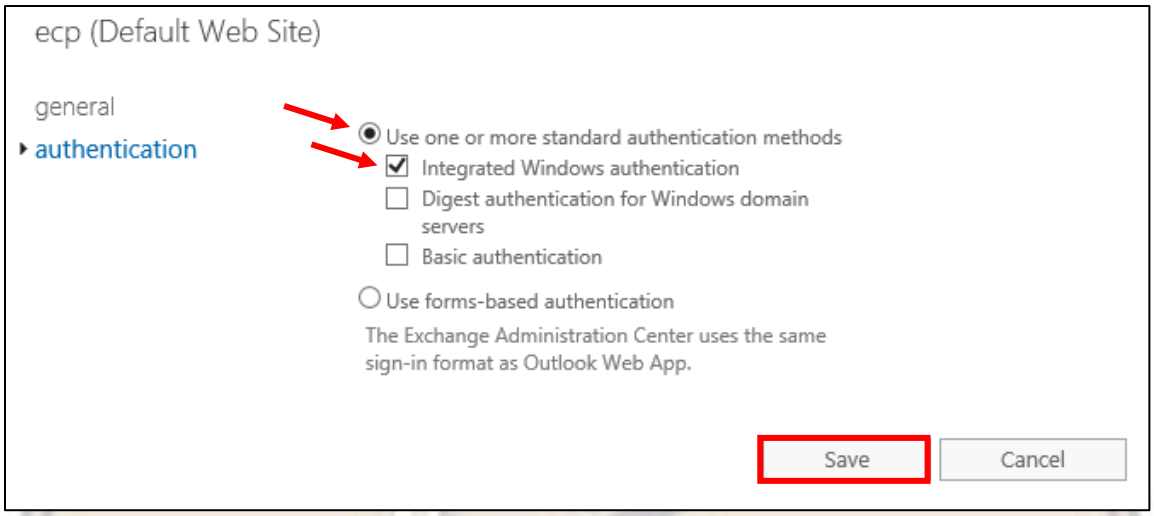
| VIRTUAL DIRECTORY | INTERNAL URL VALUE |
|---|---|
| ECP-EX01 | HTTPS://WEBMAIL.XXMEU.USMC.MIL/ECP |
| ECP-EX02 | HTTPS://WEBMAIL.XXMEU.USMC.MIL/ECP |
| OWA-EX01 | HTTPS://WEBMAIL.XXMEU.USMC.MIL/OWA |
| OWA-EX02 | HTTPS://WEBMAIL.XXMEU.USMC.MIL/OWA |
| **VIRTUAL DIRECTORY** | **EXTERNAL URL VALUE** |
| ECP-EX01 | LEAVE BLANK |
| ECP-EX02 | LEAVE BLANK |
| OWA-EX01 | HTTPS://OWA.XXMEU.USMC.MIL/OWA |
| OWA-EX02 | HTTPS://OWA.XXMEU.USMC.MIL/OWA |

3. Configure your ECP for **both Exchange Mailbox Servers** according to the chart above.

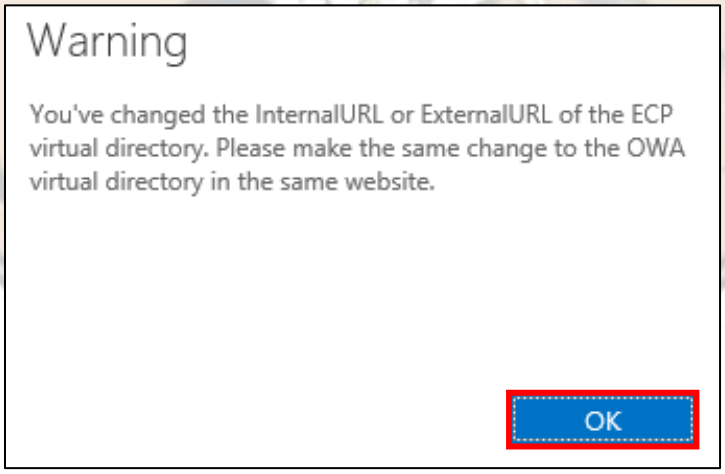
a. ECP -Change the **Internal URL** to match the chart, then click **Authentication**.



b. In Authentication, **Unselect "Forms based authentication"**, then Select **Use one or more standard authentication methods** and **Integrated Windows Authentication**. Click **Save**.

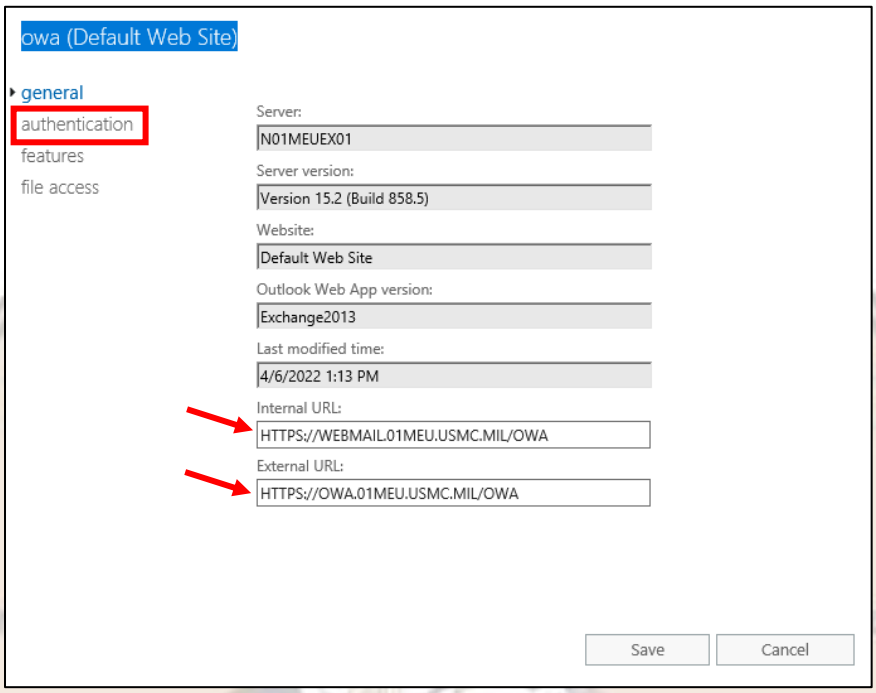



c. There is a relationship that exists with ECP and OWA, click **OK** and configure OWA as well.

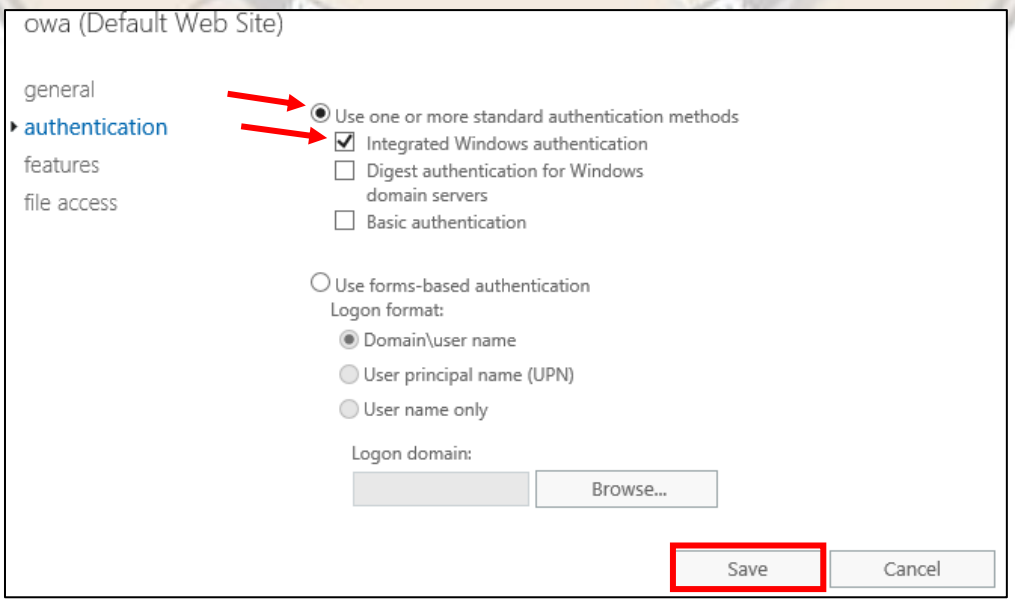


4. Configure your OWA for **both Exchange Mailbox Servers** according to the chart above.

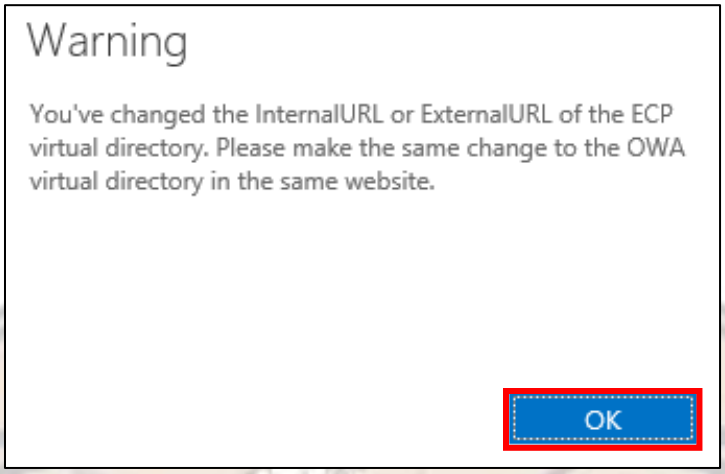
a. OWA -Change the **Internal URL** to match the chart, then click **Authentication**.




b. In Authentication, Select **<u>Use one or more standard authentication methods</u>** and **<u>Integrated Windows Authentication</u>**. Click **Save**.



c. There is a relationship that exists with ECP and OWA, click **OK** and configure OWA as well.




## Task Six: Database Availability Groups

**Note: additional admin account permissions are required (Organization Management).**

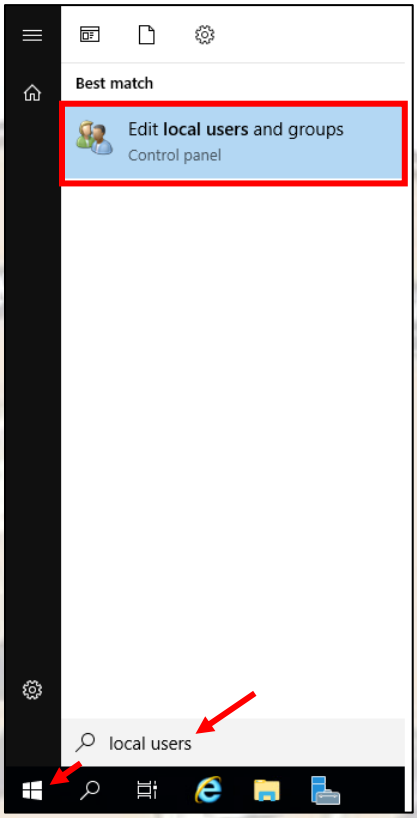
A database availability group (DAG) is the base component of the Mailbox server high availability and site resilience framework built into Microsoft Exchange Server. A DAG is a group of up to 16 Mailbox servers that hosts a set of databases and provides automatic database-level recovery from failures that affect individual servers or databases.

Any DAG Server can host a copy of a mailbox database from any other server within the DAG. When a DAG is initially created, a failover cluster, and an empty object stored in AD is created with infrastructure that monitors the servers for network or server failures using a cluster heartbeat mechanism and a cluster database to track and manage information about the DAG. A server that is added to a DAG works with the other servers to provide automatic recovery. For our purposes we will create a DAG with Two Exchange servers and The Utility server will be used as the witness.

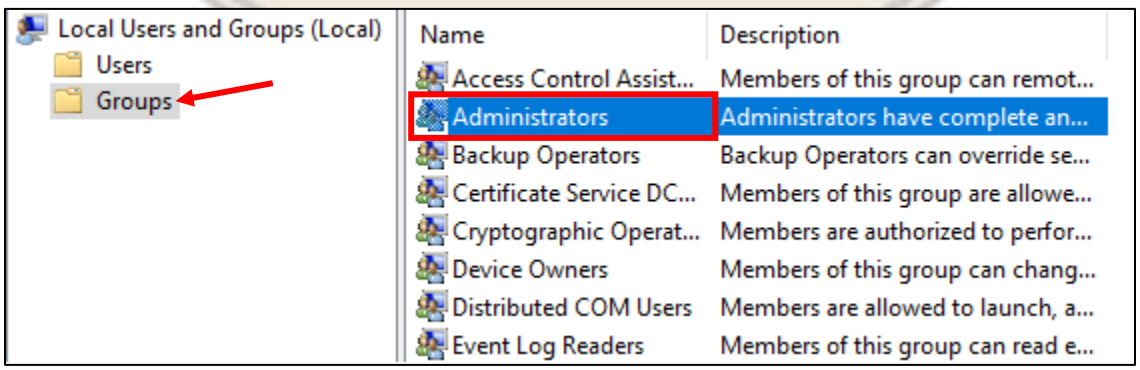
## 1. DAG Creation:

A. Prepare the Witness Host (NXXMEUUTIL)

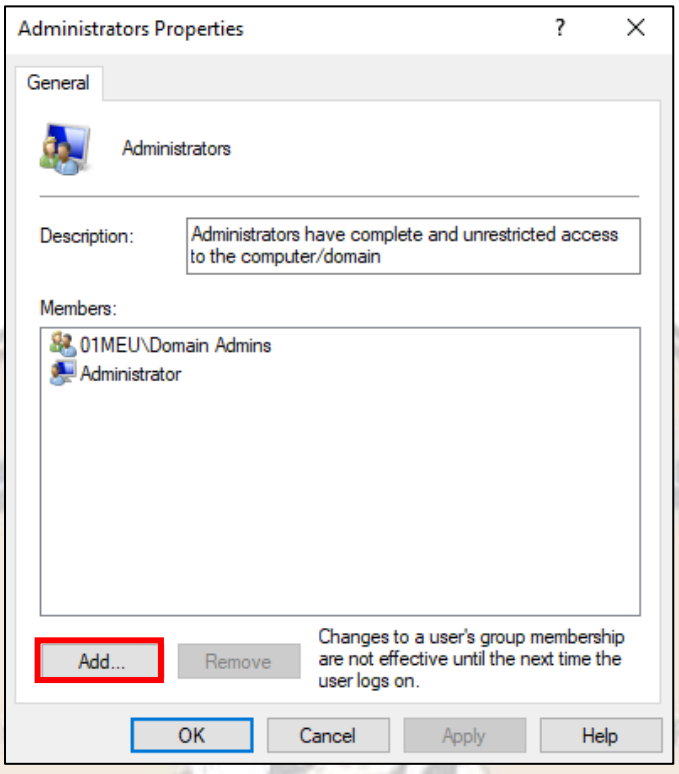
1. Login to **NXXMEUUTIL**, press the **START** key and type **local users and groups.**



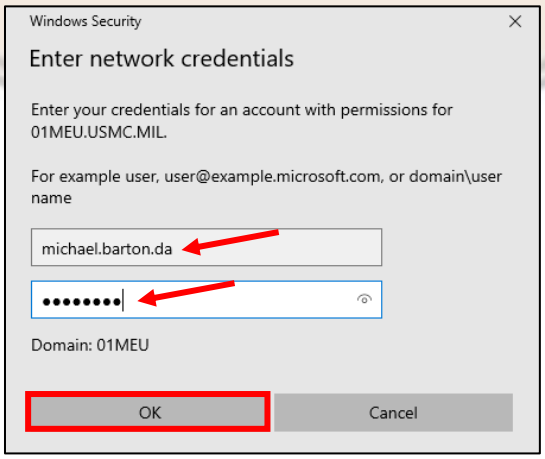

2. Open **Groups**, and double click on **Administrators**.



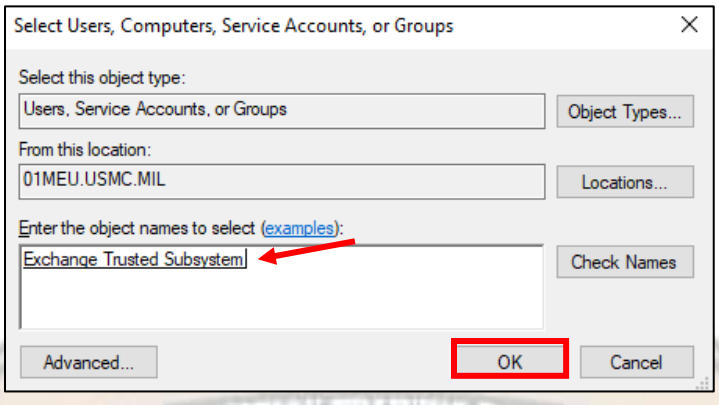
3. Click **Add**.



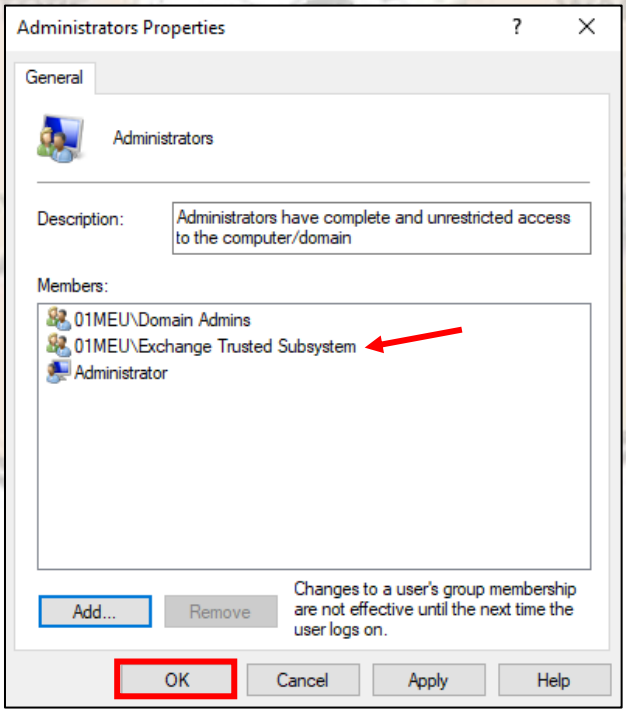

4. Enter your credentials, then Click **OK**.



6. Give the local computer **Exchange Trusted Subsystem** permissions.



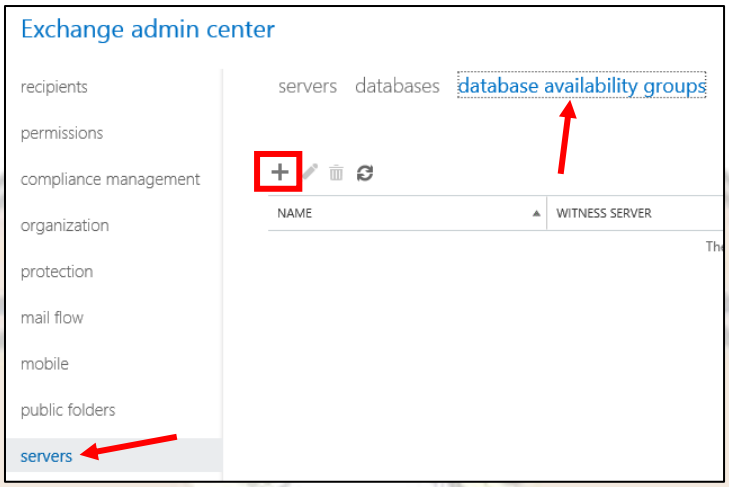

7. Click OK, then leave the NXXMEUUTIL Server.



B. **Use the EAC to create a Database Availability Group (DAG)**

We will create the DAG "**XXMEUDAG**" utilizing our **NXXMEUUTIL** server as a 'witness' server to maintain quorum.

1. In the **EAC,** go to **Servers** > **Database Availability Groups,** then **Click** + to create a **DAG**




2. On the **New Database Availability Group** page, provide the following information for the DAG:

a. DAG name: **XXMEUDAG**

b. Witness server: **NXXMEUUTIL**

**Note: If you specify a witness server, you must use either a host name or a fully qualified domain name (FQDN). Using an IP address or a wildcard name isn't supported. In addition, the witness server can't be a member of the DAG.**

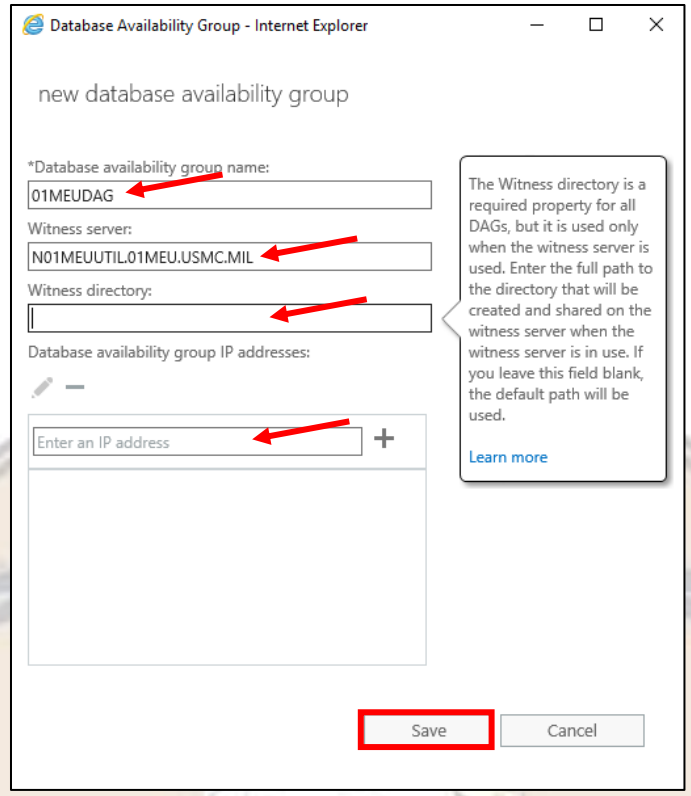
c. Witness directory: **Leave Blank the wizard will create a default directory.**

1. This will cause it to default to:

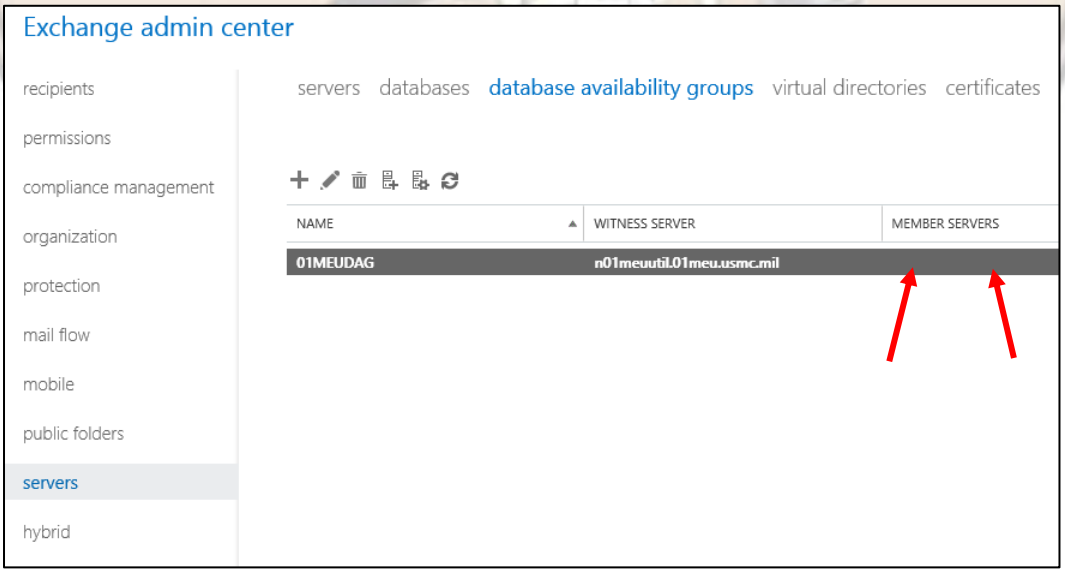
**%SystemDrive%\DAGFileShareWitnesses\<DAG FQDN>**

d. Database Availability Group IP addresses: **Leave Blank**

e. Click **Save** to **create the DAG.**

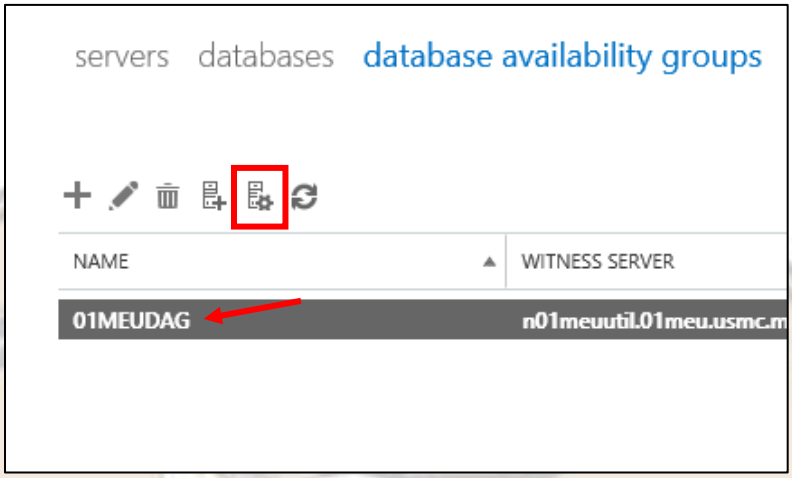



3. As you can see by default DAG's are created without members.

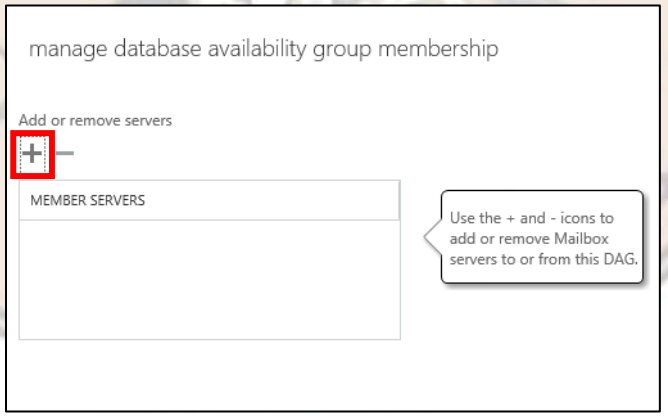



F. Add Servers to the DAG

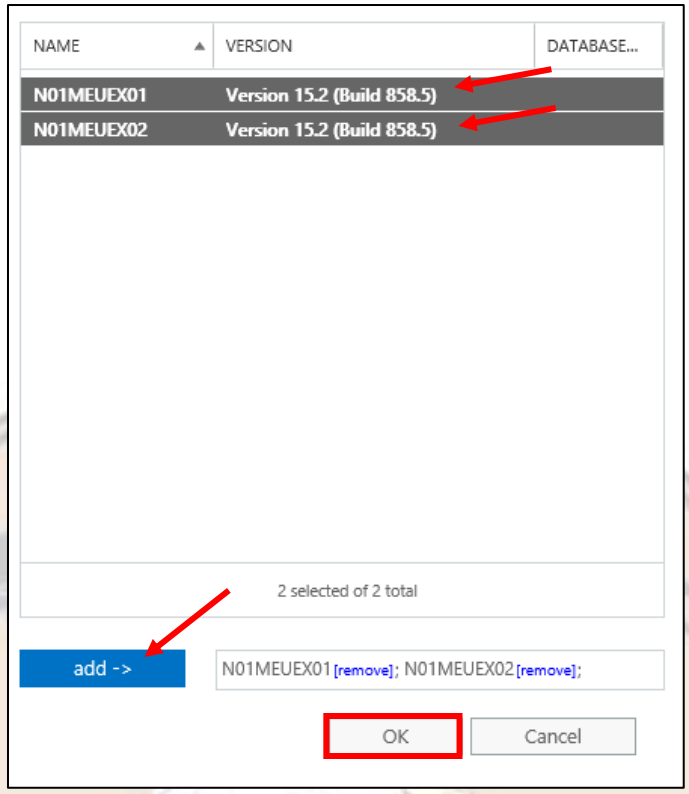
1. Under Servers → Database Availability Groups, select the DAG that you want to modify, then click  .



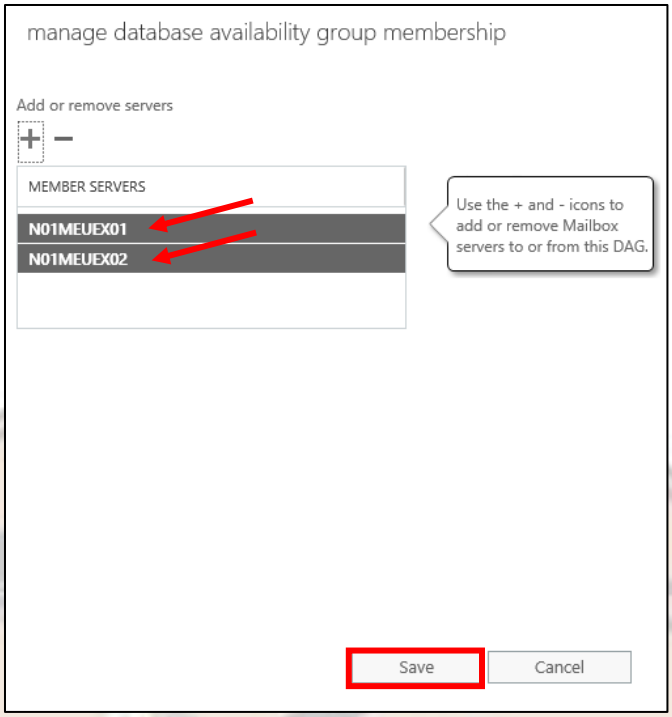

2. In the pop-up window, Select the '+'.



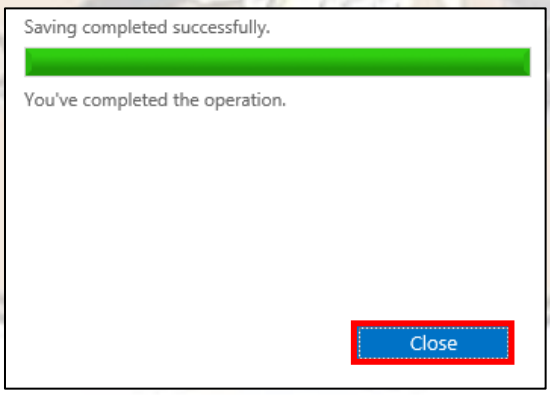

3. In the new window select both **XXMEUEX01** and **XXMEUEX02,** click Add, then Click OK.


NAME
VERSION
DATABASE...
N01MEUEX01
Version 15.2 (Build 858.5)
N01MEUEX02
Version 15.2 (Build 858.5)
2 selected of 2 total
add ->
N01MEUEX01 [remove]; N01MEUEX02 [remove];
OK
Cancel

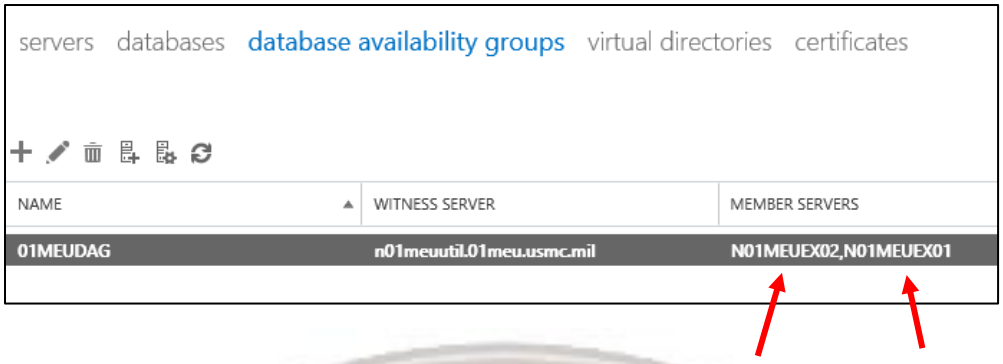
4. Click **Save**.




5. This process of forming the DAG cluster may take some time so leave the window alone. When it completes, click **Close**.




6. The Member Servers have been added.




7. If you receive any errors adding the exchange servers to the DAG, restart both Exchange servers, then try adding the servers to the DAG again.

G**. If you feel adventurous**, delete the DAG and Use the Exchange Management Shell to create a Database Availability Group.

1. The following example creates a DAG named DAG1, which is configured to use the witness server FILESRV1 and the local directory C:\DAG1. DAG1 is also configured to use DHCP for the DAG's IP addresses.

```
New-DatabaseAvailabilityGroup -Name XXMEUDAG -WitnessServer NXXMEUUTIL
   -WitnessDirectory E:\XXMEUDAG
```

H. Veryify DAG Creation

1. In the EAC, navigate to **Servers** > **Database Availability Groups**. The newly created DAG is displayed

2. In the Exchange Management Shell, run the following command to verify the DAG was created and to display DAG property information:

```
Get-DatabaseAvailabilityGroup <DAGName> | Format-List
```

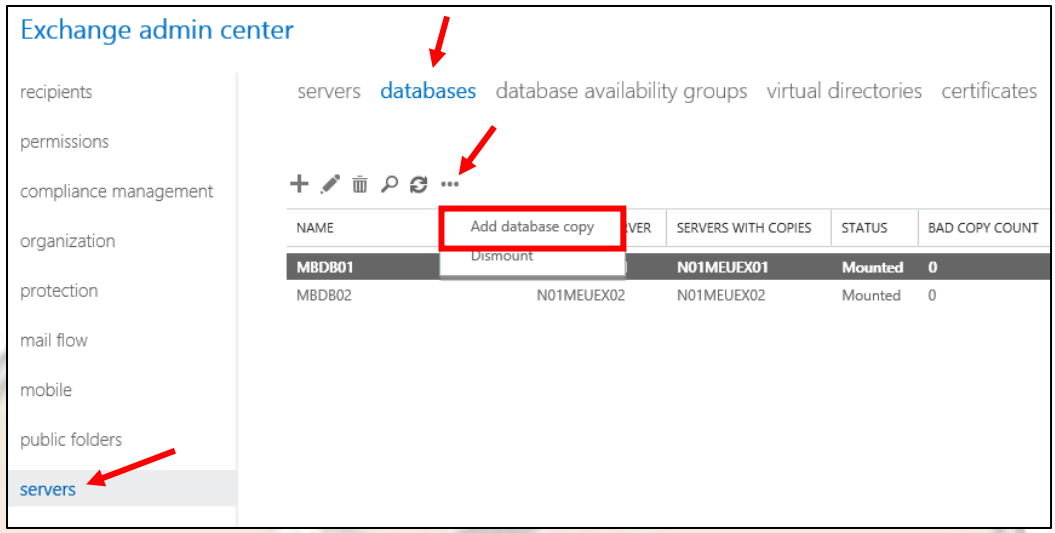
Example:

```
Get-DatabaseAvailabilityGroup XXMEUDAG | Format-List
```

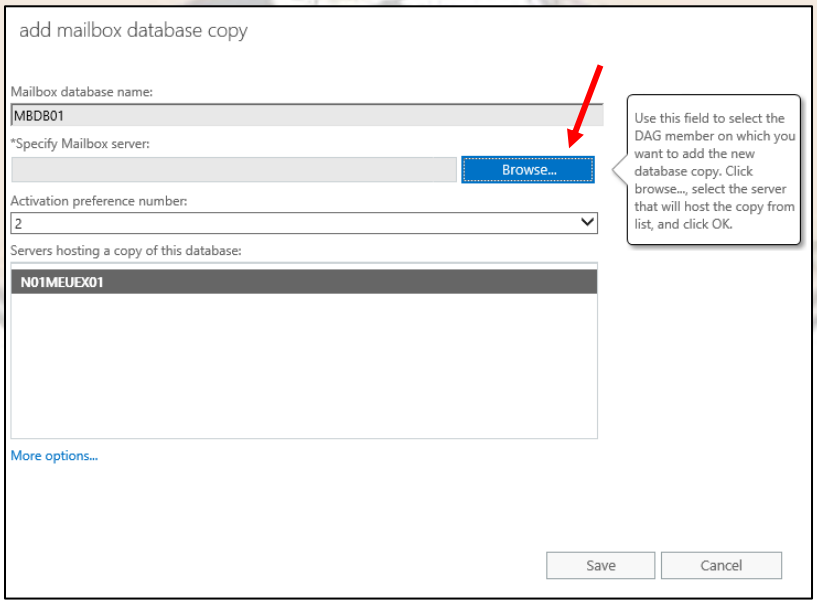
I. **Adding Database Copies**

1. For the DAG to do its job of ensuring Database level recovery, it needs to have Database Copies to '**seed**' or update. In this example, we will be creating passive copies for MBDB01 on MBDB02, and MBDB02 on MBDB01.
2. Creating a Database copy in EAC:

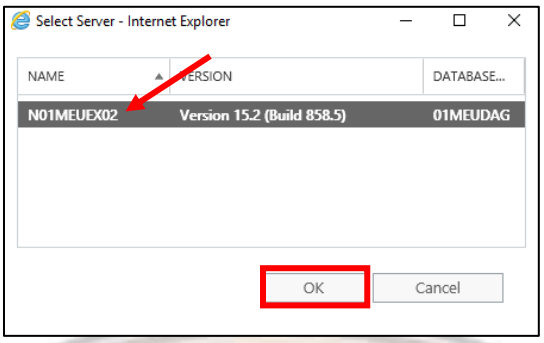
a. Navigate to **Servers → Databases**, then select the first Database you wish to copy. Click the three dots in the command row above the columns and select **Add Database Copy**.



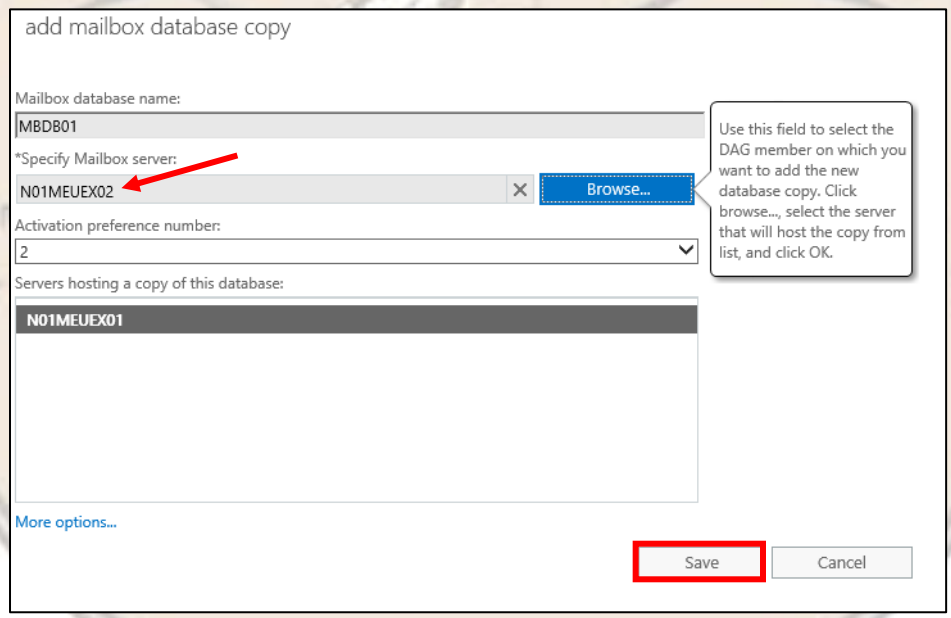

b. Click on **Browse.**




c. Select the available Mailbox Database Servers, then Click OK

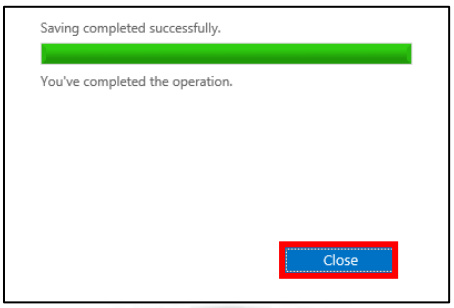



d. Click on **Save** and wait for exchange to start the seeding process.

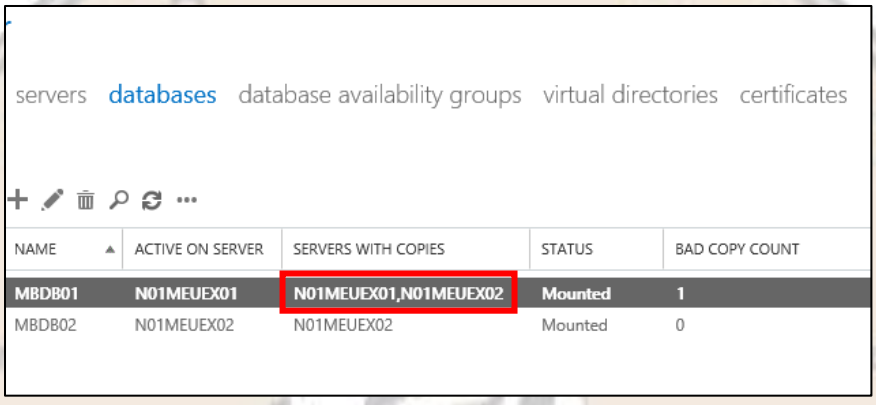



Note: You CANNOT select the witness to host a database copy and it should not be an option. In environments where you have low bandwidth, you may need to restrict your backups to **on command only.** You would need to select '**postpone seeding**' under **more options**.

e. When complete, click **Close**.

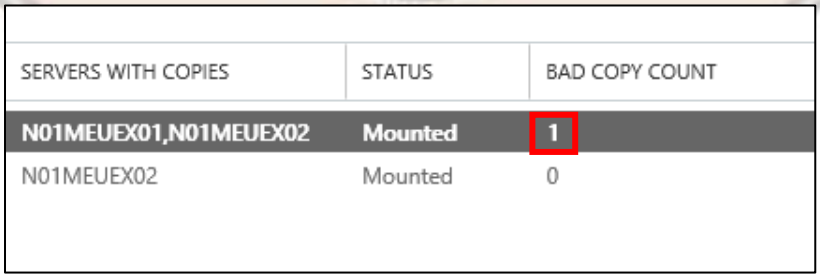



f. The Database for Exchange01 is now hosted on both servers.

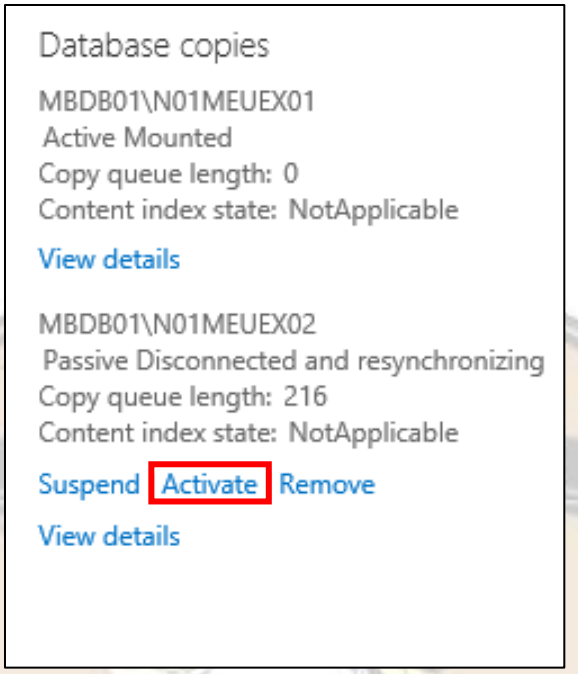



g. Repeat the steps for the other Mailbox Servers. When you are complete, you should see that both server names appear under '**Servers With Copies**'

3. Fix a Bad Copy Count




a. Click Activate to restart the seeding process.




**This Concludes Lab 2, Proceed to Lab 3.**